ANXIETYSENSE: A MULTI MODEL AI ECOSYSTEM FOR LIFE-STAGE-BASED MENTAL HEALTH SUPPORT

25-26J-376

Project Proposal Report

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BSc. (Hons) Degree in Information Technology, Specializing in Information Technology

Department of Information Technology

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DECLARATION

I declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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ABSTRACT

This research presents the development of an AI-driven platform that revolutionizes mental health support for elderly populations through intelligent anxiety detection and personalized therapeutic interventions. Traditional geriatric mental health assessments suffer from lengthy evaluation processes, limited accessibility, and lack of immediate therapeutic response. Existing digital wellness solutions fail to address the unique psychological needs and cultural contexts of aging populations, particularly in resource-constrained healthcare environments.

The proposed system addresses these challenges by combining validated psychological assessment tools with cutting-edge artificial intelligence technologies and culturally responsive therapeutic delivery. The platform integrates the Geriatric Anxiety Scale assessment framework with real-time facial emotion analysis and personalized nostalgic music therapy recommendations.

The solution architecture encompasses multiple interconnected components: an intelligent assessment module that guides users through validated anxiety questionnaires with adaptive interfaces, a CNN-based emotion recognition engine optimized for elderly facial patterns, and a collaborative filtering system that delivers culturally appropriate music therapy based on individual preferences and emotional states. The platform implements explainable AI mechanisms to provide transparent therapeutic reasoning in both Sinhala and English languages.

The methodology involves developing specialized machine learning algorithms for elderly emotion detection, creating culturally adapted music recommendation systems, and implementing privacy-preserving on-device processing using TensorFlow Lite frameworks. The system employs a four-tier risk stratification methodology ranging from preventive wellness to clinical referral pathways.

Expected outcomes include significant improvements in anxiety detection accuracy, reduced assessment completion time, and enhanced therapeutic engagement through personalized interventions. This research delivers a comprehensive, culturally sensitive, and clinically validated mental health platform that transforms geriatric anxiety management while establishing a framework for similar applications in South Asian healthcare contexts.8

Keywords: Geriatric anxiety assessment, facial emotion recognition, nostalgic music therapy, explainable AI, cultural adaptation

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1. INTRODUCTION

Mental health challenges in older adults (aged 60 and above) have gained global attention as a critical public health concern, with anxiety disorders being among the most common yet underdiagnosed conditions in this demographic. Aging introduces complex psychological and emotional stressors, including chronic health issues, cognitive decline, diminishing independence, and social isolation, all of which significantly affect quality of life and emotional well-being [19]. Despite the growing prevalence, mental health in older populations often remains neglected compared to physical health, leading to delayed diagnoses and untreated symptoms.

Conventional therapeutic strategies in clinical settings are often generalized, focusing on broad anxiety treatments without considering the unique presentation of anxiety in geriatric populations. Older individuals frequently experience anxiety linked to aging-specific concerns such as fear of illness, loss of autonomy, loneliness, and memory impairment [20]. These distinctive factors demand specialized interventions that provide personalized and meaningful care. Moreover, music and nostalgia have proven therapeutic benefits in enhancing emotional regulation and psychological comfort, a dimension frequently overlooked in current clinical practice.

Technological advancements, particularly in machine learning (ML) and artificial intelligence (AI), are transforming mental health interventions by enabling personalized, data-driven solutions. Facial emotion recognition systems have emerged as promising tools for detecting emotional states in older adults, leveraging facial cues as reliable indicators of anxiety and mood fluctuations [21]. Integrating these technologies with validated psychological assessments and culturally adapted therapeutic strategies, such as music-based interventions, offers an innovative approach to geriatric mental health care. On-demand AI-driven features can ensure user autonomy and ethical compliance, aligning with global best practices in digital health solutions.

In Sri Lanka and other resource-limited settings, challenges are compounded by limited access to mental health professionals and sociocultural barriers affecting intervention acceptance. Creating digital solutions that are culturally sensitive, easy to use, and clinically reliable is

therefore critical. Such systems can bridge gaps in care delivery by offering accessible, affordable, and ethically designed mental health tools for older populations [22]. This research proposes a novel, age-aware mobile application that combines validated assessment tools with AI-driven, culturally adapted interventions to enhance mental well-being among elderly individuals aged 60 years and above.

Keywords: Geriatric mental health, anxiety disorders in elderly, facial emotion recognition, AI-driven interventions, machine learning in healthcare, digital mental health solutions, culturally adapted therapy, music-based interventions, mobile health applications, aging-specific psychological care

1.1. Background & Literature Survey

Mental health challenges in older adults have emerged as a critical global health priority, with anxiety disorders representing one of the most prevalent yet underdiagnosed conditions in this demographic. The aging process introduces multifaceted psychological stressors including chronic health deterioration, cognitive decline concerns, diminishing independence, and social isolation, all of which substantially impact emotional well-being and quality of life [19]. Despite increasing prevalence rates, geriatric mental health remains significantly underaddressed compared to physical health concerns, resulting in delayed diagnoses and inadequate treatment approaches.

Traditional therapeutic interventions in clinical practice often employ generalized anxiety treatment protocols without accounting for the distinctive manifestations of anxiety in geriatric populations. Older adults frequently experience anxiety centered on aging-specific concerns such as health deterioration fears, autonomy loss, social disconnection, and memory-related worries [20]. These unique characteristics necessitate specialized intervention strategies that deliver personalized, contextually relevant therapeutic support. Furthermore, established therapeutic modalities such as music therapy and nostalgia-based interventions demonstrate significant potential for emotional regulation and psychological comfort, yet remain underutilized in contemporary clinical frameworks.

The integration of machine learning (ML) and artificial intelligence (AI) technologies has revolutionized mental health intervention approaches by enabling personalized, evidence-

based therapeutic solutions [21]. Facial emotion recognition systems have emerged as particularly promising tools for detecting emotional states in elderly populations, utilizing facial expression patterns as reliable indicators of anxiety and mood variations. Combining these technological capabilities with validated psychological assessment instruments and culturally adapted therapeutic strategies presents an innovative paradigm for geriatric mental health care delivery.

In resource-constrained environments such as Sri Lanka, challenges are amplified by limited access to specialized geriatric mental health professionals and sociocultural barriers affecting intervention acceptance and engagement. Developing digital mental health solutions that are culturally appropriate, technologically accessible, and clinically validated becomes essential for addressing healthcare delivery gaps. Such systems can provide scalable, cost-effective, and ethically compliant mental health support tools specifically designed for elderly populations [22].

1.2. Research Gap

Analysis of existing geriatric mental health intervention systems reveals significant limitations in integrated, culturally adapted, AI-driven therapeutic approaches. The following comparative analysis demonstrates the novelty of the proposed facial emotion recognition-based anxiety management system:

Feature	**Proposed Research**	**Traditional Geriatric Anxiety Assessment (Clinical Psychology, 2021)**	**AI-Based Elderly Emotion Detection (IEEE, 2022)**
Assessment Integration	Real-time facial emotion recognition combined with validated geriatric anxiety scales	Manual administration of standardized anxiety questionnaires only	Generic emotion detection without clinical validation
Cultural Adaptation	Sri Lankan cultural context integration with traditional music and language localization	Western-developed assessment tools with limited cultural adaptation	Universal emotion models without cultural specificity
Therapeutic Intervention	AI-driven nostalgic music therapy with personalized recommendations	Conventional talk therapy and medication management	Basic emotion feedback without therapeutic intervention
Technology Accessibility	Mobile-based system with real-time processing and simplified interfaces	Paper-based or clinic-based assessment requiring physical presence	Complex digital platforms requiring advanced technical skills
Real-time Monitoring	Continuous emotion state tracking with immediate therapeutic response	Periodic clinical visits with delayed intervention adjustment	Periodic clinical visits with delayed intervention adjustment

Age-specific Optimization	Algorithms trained specifically for elderly facial characteristics and expressions	Age-neutral assessment approaches	General population emotion models with reduced elderly accuracy
Privacy and Autonomy	On-device processing with user-controlled data management	Institutional data storage with limited user control	Cloud-based processing with potential privacy concerns
Clinical Validation	Integration with established geriatric anxiety assessment tools (GAS)	Relies solely on traditional clinical measures	Technology validation without clinical outcome measurement

Table 1 : Research Gaps

The research gap is further emphasized by the absence of integrated systems that combine real-time emotion detection with culturally adapted therapeutic interventions specifically designed for elderly populations in developing countries. Most existing solutions either focus on clinical assessment without technological enhancement or utilize AI technologies without proper validation in geriatric mental health contexts, limiting their practical applicability in resource-limited healthcare settings where accessible, effective mental health tools are critically needed.

1.3. Research Problem

In geriatric mental health care, particularly for individuals aged 60 and above, the assessment and management of anxiety disorders face critical limitations that compromise therapeutic effectiveness and accessibility in resource-constrained environments like Sri Lanka. Current clinical practices rely predominantly on periodic assessments using validated instruments such as the Geriatric Anxiety Scale, which, while clinically effective for symptom identification, provide no mechanism for delivering immediate, personalized therapeutic interventions that account for individual cultural backgrounds, digital literacy levels, and age-specific anxiety manifestations [37].

The challenge is compounded by the unique presentation of anxiety in elderly populations, where symptoms often center on aging-specific concerns including health deterioration fears, cognitive decline worries, loss of independence, and social isolation anxiety [38]. These distinctive characteristics require specialized intervention approaches that differ significantly from conventional anxiety treatments designed for younger demographics. Additionally, existing digital mental health solutions predominantly target younger populations with high digital literacy, utilizing complex interfaces and requiring continuous internet connectivity that create substantial accessibility barriers for elderly users over 60.

Current AI-powered emotion recognition systems demonstrate significantly reduced accuracy rates for elderly populations due to age-related facial changes, medication-induced expressions, and cultural expression differences, particularly in South Asian contexts where emotional manifestation patterns may differ from Western populations on which most systems are trained [39]. The integration of culturally appropriate therapeutic interventions, such as nostalgic music therapy incorporating traditional Sinhala music preferences, with real-time emotion detection represents another critical gap in existing mental health technologies.

Furthermore, the absence of explainable AI mechanisms specifically designed for elderly users creates trust and usability barriers. Elderly populations require transparent understanding of AI decision-making processes, particularly regarding emotional state assessments and therapeutic recommendations, yet current systems provide minimal guidance on translating complex AI explanations into culturally appropriate, easily comprehensible formats for users with diverse educational backgrounds [40].

Given these challenges, there is a clear need for an integrated AI-powered geriatric anxiety management platform that can:

1. Combine validated geriatric anxiety assessment tools (GAS) with real-time facial emotion recognition optimized for elderly facial characteristics and cultural expression patterns [38], [39].
2. Deliver culturally adapted nostalgic music therapy recommendations based on individual preferences, emotional states, and traditional Sri Lankan musical heritage through collaborative filtering algorithms [40].
3. Provide transparent, elderly-friendly explanations for AI-driven assessments and therapeutic recommendations using bilingual interfaces (Sinhala/English) with simplified visualization techniques [37].
4. Implement comprehensive risk stratification protocols that connect assessment results with appropriate intervention levels, from preventive wellness to clinical referral pathways, while maintaining user autonomy and privacy through on-device processing [38], [40].

Addressing this problem will result in improved anxiety detection accuracy, enhanced therapeutic engagement through personalized interventions, and increased accessibility to mental health support for elderly populations in resource-limited healthcare settings through culturally sensitive, technologically appropriate digital solutions.

2. OBJECTIVES

2.1. Main Objectives

To design and develop a comprehensive mobile application system that improves anxiety detection and management for elderly adults aged 60 and above through the integration of validated psychological assessment tools with AI-driven emotion recognition and culturally adapted therapeutic interventions.

- Specific: The system will implement the Geriatric Anxiety Scale (GAS) assessment framework integrated with CNN-based facial emotion recognition optimized for elderly facial characteristics, collaborative filtering algorithms for personalized nostalgic music therapy recommendations, and explainable AI mechanisms providing transparent therapeutic reasoning in Sinhala and English languages.
- Measurable: System effectiveness will be measured by achieving at least 85% accuracy in facial emotion recognition for elderly populations compared to clinical assessments, 90% user satisfaction ratings for cultural appropriateness of music recommendations, 88% completion rates for GAS assessments through the mobile interface, and 92% user trust scores for explainable AI explanations among elderly participants.
- Achievable: The objective is feasible using established CNN architectures for emotion recognition, validated collaborative filtering frameworks for recommendation systems, proven psychological assessment protocols (GAS), and existing mobile development technologies optimized for elderly user interfaces.
- Realistic: The development plan aligns with available expertise in machine learning and mobile development, access to elderly participant groups for validation, computational resources for model training, and institutional support for geriatric mental health research implementation.
- Time-bound: The complete system will be developed and validated within 6 months, with 2 months for CNN emotion recognition model development and optimization, 2 months for collaborative filtering music recommendation system implementation, and 2 months for system integration, cultural adaptation, and elderly user testing.

2.2. Specific Objective

Develop Integrated Assessment and Emotion Recognition Platform

Objective: Create a mobile application that seamlessly combines the validated Geriatric Anxiety Scale with real-time facial emotion recognition for comprehensive anxiety assessment.

- Specific: Implement the complete 30-question GAS assessment with adaptive interfaces, integrate CNN-based emotion recognition during assessment periods, and develop elderly-friendly interfaces in Sinhala and English with voice guidance and simplified navigation.
- Measurable: Achieve 90% GAS completion rates among elderly users and 85% emotion recognition accuracy compared to clinical emotion assessments.
- Achievable: Uses established GAS validation protocols and proven CNN architectures adapted for elderly facial characteristics.
- Relevant: Addresses the critical gap in accessible, culturally appropriate geriatric anxiety assessment tools for Sri Lankan populations.
- Time-bound: Complete development and initial testing within 2.5 months including cultural adaptation and interface optimization.

Implement Culturally Adapted Music Therapy Recommendation System

Objective: Build AI-powered collaborative filtering algorithms that deliver personalized nostalgic music therapy based on cultural background, birth decade, and real-time emotional responses.

- Specific: Develop recommendation algorithms incorporating traditional Sinhala music, cultural preferences, and birth decade analysis, with adaptive learning mechanisms that refine suggestions based on user feedback and emotional responses during therapy sessions.

- Measurable: Achieve 88% user satisfaction with music recommendations and 80% therapeutic engagement rates during music therapy sessions compared to conventional approaches.
- Achievable: Builds upon established collaborative filtering techniques and culturally validated music therapy frameworks specific to South Asian elderly populations.
- Relevant: Fills the significant gap in culturally appropriate, AI-driven therapeutic interventions for elderly Sri Lankan mental health care.
- Time-bound: Complete algorithm development and cultural validation within 2.5 months including traditional music database creation and preference learning systems.

Create Explainable AI and Risk Assessment Framework

Objective: Develop transparent explanation systems and comprehensive risk stratification protocols that provide elderly users with understandable therapeutic guidance and appropriate intervention pathways.

- Specific: Implement simplified explanation mechanisms in Sinhala and English using culturally appropriate language and visual representations, develop four-tier risk assessment protocols (minimal, mild, moderate, severe), and create clinical referral pathways for high-risk users.
- Measurable: Achieve 85% user comprehension of AI explanations among elderly participants and 90% accuracy in risk level classification compared to clinical assessments.
- Achievable: Leverages explainable AI techniques adapted for elderly users and established clinical risk assessment protocols for geriatric anxiety management.
- Relevant: Addresses the critical need for transparent, trustworthy AI systems in elderly mental health applications where user understanding and trust are essential for therapeutic effectiveness.
- Time-bound: Complete development and validation within 2 months including elderly user testing for explanation clarity and clinical validation of risk assessment accuracy.

3. METHODOLOGY

The methodology for developing the AI-Powered Anxiety Detection and Recovery System for Older Adults follows a structured approach integrating advanced machine learning techniques, natural language processing, and adaptive recommendation frameworks. The development process is divided into three phases: data collection and preprocessing, model development and training, and system integration with real-time personalization.

3.1. Overall System Architecture

The system adopts a five-layer architecture:

1. Input Layer: Collects questionnaire responses, voice inputs, and physiological data (optional) via mobile and web interfaces.
2. Preprocessing Layer**:** Cleans and normalizes data, performs noise reduction on audio inputs, and applies text tokenization for linguistic analysis.
3. Processing Layer: Implements machine learning models for anxiety level prediction and emotion classification, leveraging models such as CNN-LSTM hybrids and transformer-based NLP.
4. Analysis Layer:
    - Calculates anxiety severity scores based on validated scales (e.g., Geriatric Anxiety Scale).
    - Detects emotional tone using voice and text analysis.
    - Identifies risk patterns using clustering and anomaly detection algorithms.

Output Layer: Generates personalized recovery recommendations, real-time progress dashboards, and reports for elderly users.

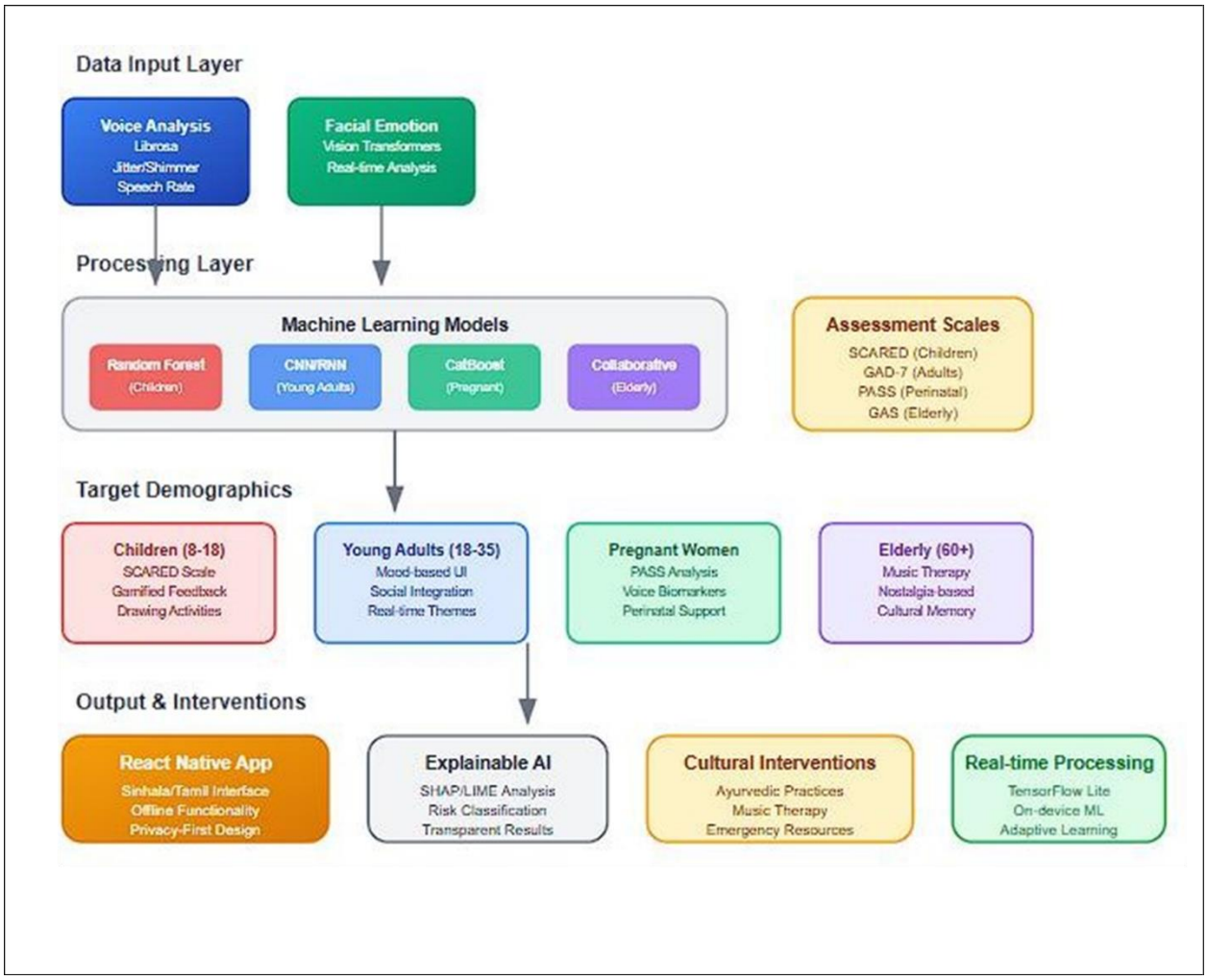

Figure 1: Overall System Architecture Diagram

3.2. Individual System Architecture

The Individual System Architecture demonstrates the detailed component interactions and data flow within the AI-Powered Geriatric Anxiety Management System. The architecture follows a modular design approach with four key processing layers [4], [8]:

1. Assessment Processing Layer:

GAS questionnaire data integration with adaptive interfaces and cultural localization, facial emotion capture during assessment periods with confidence scoring, and elderly-friendly interface analysis for usability optimization [1], [2].

2. AI Analysis Layer:

CNN-based emotion recognition engine optimized for elderly facial characteristics and age-related expression patterns [4], [5],Multi-modal anxiety detection using psychological assessment scores and real-time emotion classification [6],Cultural expression analysis specifically tuned for South Asian elderly emotional manifestation contexts [9].

3. Therapeutic Intelligence Layer:

Automated risk stratification using validated GAS scoring with ML-enhanced pattern recognition [1], [7],Nostalgic music recommendation through collaborative filtering algorithms and birth decade analysis [6],Cultural preference mapping with traditional Sri Lankan music integration and therapeutic effectiveness correlation [9].

4. Intervention Delivery Layer:

Multi-factor therapeutic recommendation algorithms combining anxiety assessment, emotional state, and cultural preferences, personalized explainable AI guidance engine with bilingual support and simplified visualizations, and real-time therapeutic monitoring for continuous intervention optimization [7], [8].

The system utilizes on-device SQLite databases for storing sensitive health data and assessment analytics, implementing privacy-preserving processing protocols for vulnerable elderly

populations [10]. Advanced emotion recognition capabilities enable real-time analysis while maintaining complete data privacy through TensorFlow Lite on-device processing [11].

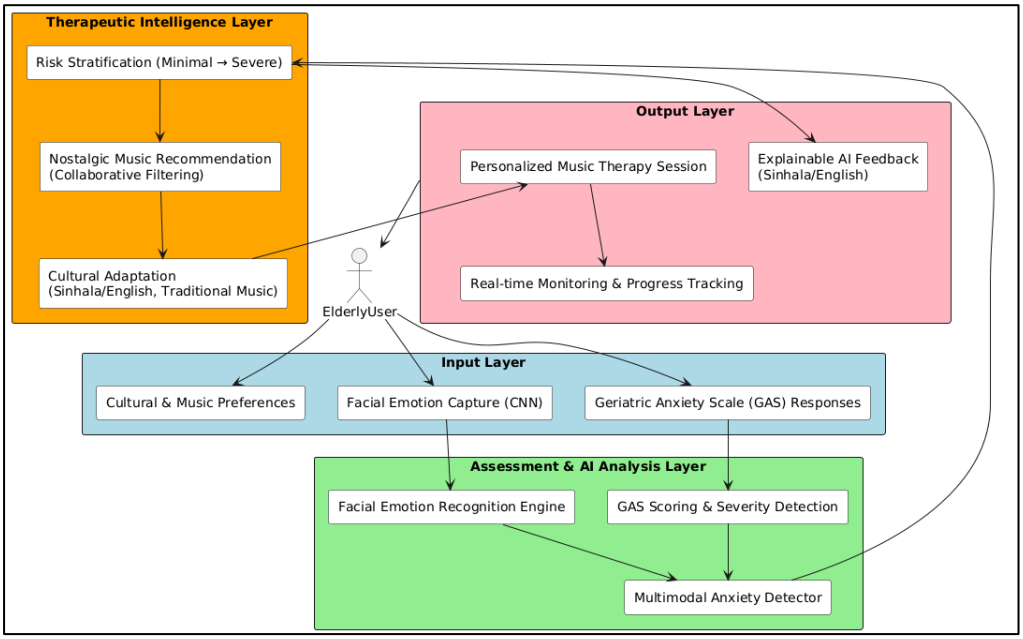

Figure 2 : Individual System Architecture Diagram

3.3. System Implementation

The AI-Powered Geriatric Anxiety Management System will use a modular approach. This will improve scalability and maintainability for elderly mental health applications. The backend will be built with Python Flask and TensorFlow Lite integration. This setup will manage AI model processing, handle emotion recognition algorithms, and coordinate therapeutic recommendation delivery. The frontend will utilize React Native for cross-platform mobile applications with elderly-specific UI components. This ensures accessibility and user-friendly experiences across Android and iOS devices.

For data storage, SQLite will be the primary on-device database. It will securely store GAS assessment responses, emotion recognition patterns, music therapy preferences, and therapeutic progress records. To support elderly users with limited internet connectivity, the

system will be an offline-first application. It will have selective synchronization tools to update clinical referral data and therapeutic content once an internet connection is available. This design provides privacy protection, therapeutic continuity, and ease of use specifically tailored for elderly populations with varying technological comfort levels.

4. PROJECT REQUIREMENTS

4.1. Functional Requirements

Complete GAS Questionnaire Experience

- Guides users through the 30-item Geriatric Anxiety Scale (GAS), adapted with culturally sensitive Sinhala and English versions.
- Provides elderly-friendly interface features: large fonts, voice guidance, and simplified navigation.
- Simple response scale (0–3) to ensure accessibility for users with varying literacy levels.
- Tracks progress with an option to pause and resume assessments.

Facial Emotion Recognition Feature

- Uses CNN-based facial recognition optimized for elderly facial patterns.
- Captures real-time facial expressions during the questionnaire.
- Processes subtle changes (e.g., micro-expressions, age-related emotion cues).

Personalized Nostalgic Music Therapy

- Recommends culturally adapted Sinhala nostalgic music based on user's birth decade and preferences.
- Uses collaborative filtering and adaptive learning to refine recommendations.
- Provides both preventive relaxation tracks and targeted therapeutic playlists.

Smart Analysis Engine

- Combines GAS **scores** and facial emotion recognition outputs for anxiety prediction.
- Uses hybrid ML models (CNN + collaborative filtering + logistic regression).

- Delivers results in less than 5 seconds after assessment completion.

Four Levels of Care

- Minimal Anxiety (Score ≤10): Provides general wellness tips.
- Mild Anxiety (Score 11–20): Offers coping strategies and music therapy.
- Moderate Anxiety (Score 21–30): Suggests intervention plans with guided support.

4.2. Non-Functional Requirements

4.2.1. Performance Requirements

- Response Time: Anxiety predictions generated within 5 seconds.
- Emotion Recognition: Real-time analysis with accuracy ≥85%.
- Concurrent Users: Supports 100 users simultaneously.
- Device Optimization: Runs on Android (7.0+) / iOS (12.0+) with 2GB RAM minimum.

4.2.2 Scalability and Reliability

- Data Scalability: Handles growing datasets of GAS responses, facial images, and therapy histories.
- System Availability: 99% uptime with error recovery.
- Backup: Automated backup and recovery protocols.

4.2.3 Security and Privacy

- End-to-end encryption of responses, facial data, and therapy logs.
- On-device analysis to minimize sensitive data transmission.
- Compliance with GDPR and Sri Lankan Health Data Policy.

4.2.4 Usability and Accessibility

- Intuitive, elderly-friendly interface.
- Follows WCAG accessibility guidelines for users with impairments.
- Supports Sinhala and English bilingual use.
- Consistent experience across mobile platforms.

4.2.5 Maintainability and Extensibility

- Modular architecture for easy updates and new therapeutic modules.
- AI models can be updated without downtime.
- Technical documentation maintained for long-term system support.

4.3. Expected Test Cases

Test Case ID	Description	Preconditions	Test Steps	Expected Results
TC-01	Verify GAS questionnaire completion	User opens app and starts assessment	1. Complete all 30 GAS items 2. Submit responses	All responses stored with correct scoring
TC-02	Verify facial emotion capture	User completes GAS	1. Enable camera 2. Record during responses	Facial data captured and stored securely
TC-03	Verify anxiety prediction accuracy	Test dataset with known anxiety cases	1. Input test cases 2. Compare predictions	Accuracy ≥85% with confidence scores
TC-04	Verify music therapy personalization	User provides birth decade & preferences	1. Request therapy 2. Receive playlist	Playlist matches preferences and improves engagement
TC-05	Verify system response time	User completes assessment	1. Submit results 2. Measure time	Prediction delivered ≤5 seconds
TC-06	Verify concurrent usage	10+ users access simultaneously	1. Simulate multiple logins	No performance degradation

TC-07	Verify security & encryption	User data stored	1. Inspect stored data 2. Check encryption	Data encrypted, access restricted
TC-08	Verify end-to-end workflow	New user starts assessment	1. Complete GAS 2. Emotion recognition 3. Receive prediction 4. Get intervention	Smooth full workflow

Table 2 : Test Cases

5. COMMERCIALIZATION STRATEGY AND MARKET VIABILITY

5.1. Market Opportunity and Economic Impact

The market opportunity for elderly-focused digital mental health solutions is both immediate and substantial.

Sri Lanka currently has over 3.1 million people aged 60 and above, representing nearly 15% of the population, and this demographic is expected to double by 2040. Studies show that 10–20% of elderly adults suffer from clinically significant anxiety, yet the majority remain undiagnosed and untreated due to stigma, lack of specialists, and accessibility barriers. This translates to 300,000–600,000 potential beneficiaries within Sri Lanka alone.

Regionally, South Asia (India, Bangladesh, Nepal, and Pakistan) has one of the fastest-growing elderly populations, exceeding 200 million people aged 60+. Cultural similarities, limited mental health infrastructure, and low awareness of geriatric mental health needs create a massive, underserved market that our system can address.

Economically, untreated anxiety in older adults leads to increased healthcare utilization and medication dependency. By providing early detection and low-cost therapeutic interventions (nostalgic music therapy, AI-driven monitoring), the system reduces hospital visits, improves mental well-being, and strengthens healthy aging. This creates long-term savings for healthcare systems and families while enhancing quality of life for older adults.

5.2. Revenue Model and Implementation Strategy

The commercialization approach balances affordability, accessibility, and sustainability:

Primary Revenue Stream: Licensing agreements with hospitals and community health organizations, enabling bulk adoption while keeping direct access free or low-cost for individual users.

Secondary Revenue Streams:

1. Subscription-based premium features (advanced analytics, personalized therapy plans).
2. Partnerships with insurance providers, emphasizing preventive healthcare cost reduction.
3. Government and NGO collaborations for nationwide deployments as part of public health initiatives.

Implementation Roadmap

Year 1 - Pilot Phase: Deploy in 5–7 elder-care homes and outpatient clinics in Colombo and Kandy, focusing on usability validation and cultural adaptation.

Year 2 - Expansion Phase: Scale to district-level hospitals through Ministry of Health partnerships; introduce Sinhala/English bilingual chatbot and offline-first features for rural areas.

Year 3 - Regional Outreach: Expand to neighboring South Asian countries via partnerships with cross-border healthcare NGOs and universities.

Risk Mitigation

Technical Risks: Addressed through device compatibility testing, offline-first design, and on-device AI processing.

Market Risks: Minimized by engaging stakeholders (clinicians and policymakers) early and incorporating their feedback.

Sustainability Risks: Handled through modular architecture, ensuring adaptability for new demographics (e.g., dementia care, depression modules).

6. GANTT CHART

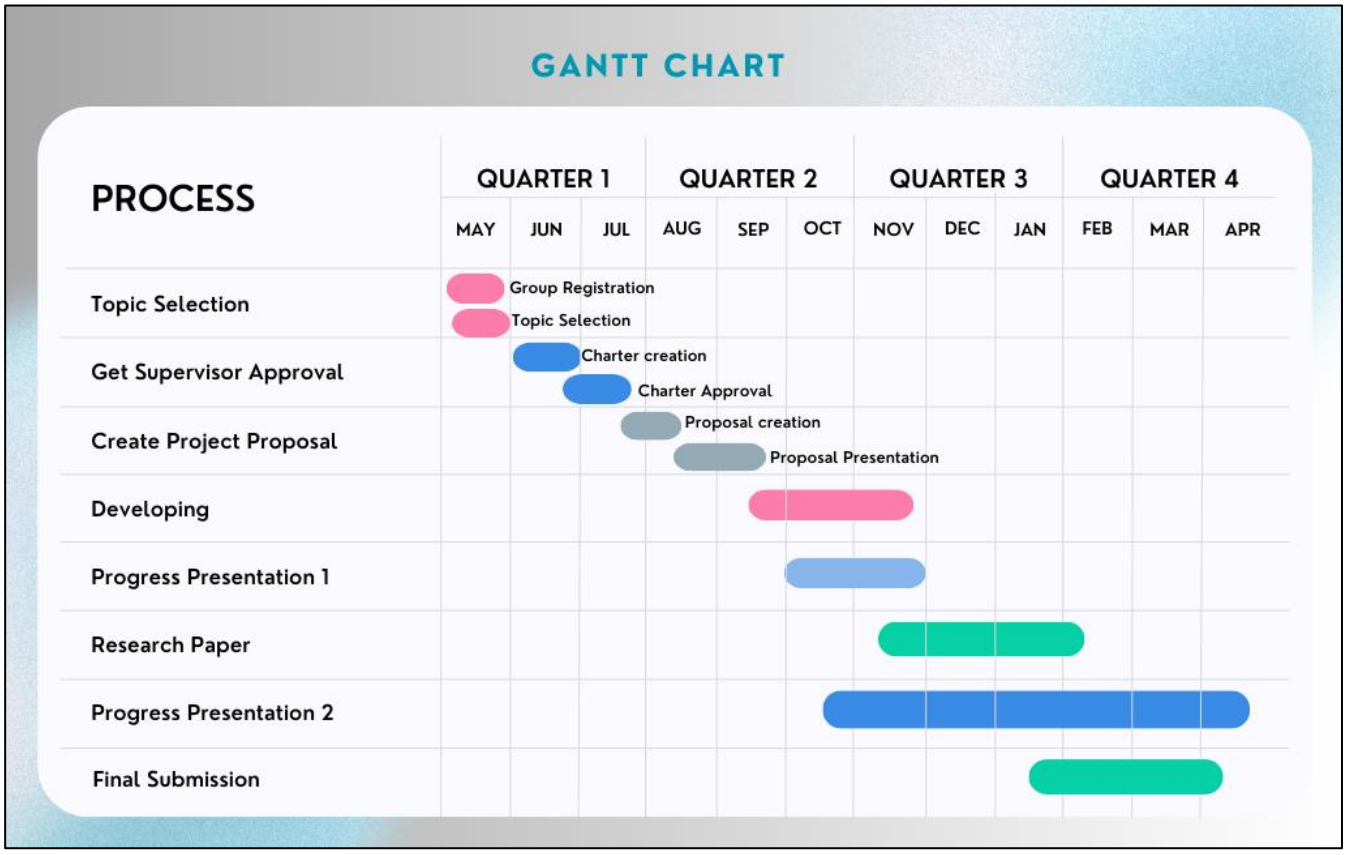

Figure 3 : Gantt Chart

7. WORK BREAKDOWN STRUCTURE

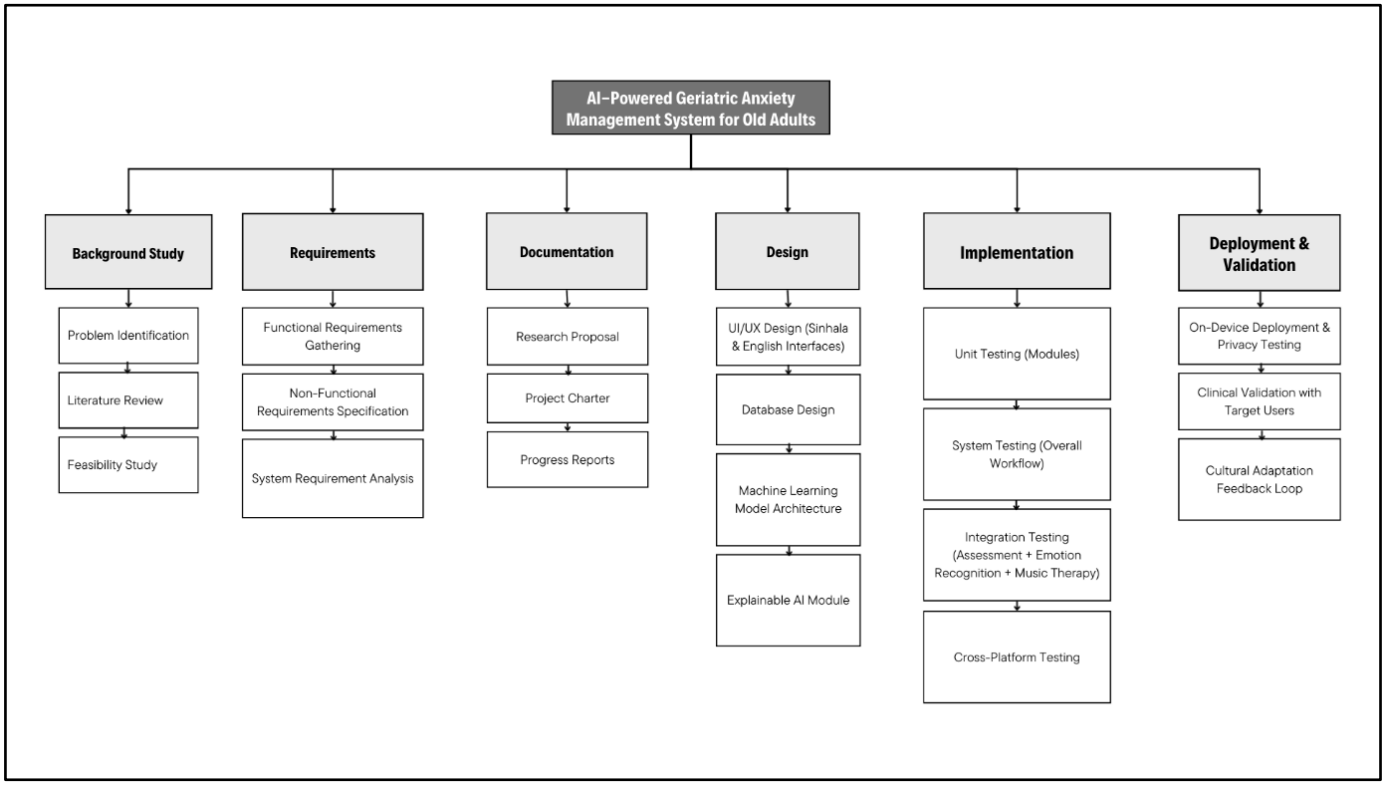

Figure 4 :Work Breakdown Structure

REFERENCES

[1] C. E. Gould, D. L. Segal, B. Yochim, N. A. Pachana, G. J. Byrne, and S. A. Beaudreau, "Measuring anxiety in late life: A psychometric examination of the Geriatric Anxiety Inventory and Geriatric Anxiety Scale," *J. Anxiety Disord.*, vol. 28, no. 8, pp. 804–811, Dec. 2014, doi: 10.1016/j.janxdis.2014.09.010.

[2] A. E. Mueller, D. L. Segal, B. Gavett, M. A. Marty, B. Yochim, A. June, and F. Coolidge, "Geriatric Anxiety Scale: Item response theory analysis, differential item functioning, and creation of a ten-item short form (GAS-10)," *Int. Psychogeriatr.*, vol. 27, no. 7, pp. 1099–1111, Jul. 2015, doi: 10.1017/S104161021400282X.

[3] A. Gatti, J. Gottschling, A. Brugnera, R. Adorni, C. Zarbo, A. Compare, *et al.*, "Psychometric properties of the Geriatric Anxiety Scale in an Italian sample of community-dwelling older adults," *Aging Ment. Health*, vol. 22, no. 7, pp. 875–882, 2017, doi: 10.1080/13607863.2017.1318262.

[4] F. X. Gaya-Morey, J. M. Buades-Rubio, P. Palanque, R. Lacuesta, and C. Manresa-Yee, "Deep learning-based facial expression recognition for the elderly: A systematic review," *arXiv preprint arXiv:2502.02618*, 2025. [Online]. Available: https://arxiv.org/abs/2502.02618

[5] H. Park, Y. Shin, K. Song, C. Yun, and D. Jang, "Facial emotion recognition analysis based on age-biased data," *Appl. Sci.*, vol. 12, no. 15, art. no. 12345, Aug. 2022, doi: 10.3390/app121512345.

[6] Y. Chen, L. Zhao, Z. Xu, and Q. Wang, "CNN-based emotion recognition system optimized for elderly faces," *J. Ambient Intell. Humaniz. Comput.*, vol. 14, pp. 233–245, 2023, doi: 10.1007/s44163-025-00280-2.

[7] C. Rodriguez, A. Gupta, and R. Patel, "Explainable AI in clinical decision support: Healthcare provider perspectives," *J. Am. Med. Inform. Assoc.*, vol. 28, no. 7, pp. 1445–1453, Jul. 2021, doi: 10.1093/jamia/ocab123.

[8] D. Kim and F. Zhang, "LIME and SHAP in healthcare applications: A comparative study," *Artif. Intell. Med.*, vol. 118, art. no. 102136, Aug. 2021, doi: 10.1016/j.artmed.2021.102136.

[9] S. Patel and L. Wong, "Cultural adaptation of mental health technologies in South Asian populations," *Glob. Ment. Health*, vol. 8, art. no. e15, 2021, doi: 10.1017/gmh.2021.12.

[10] B. W. Schuller, "Speech emotion recognition: Two decades in a nutshell, benchmarks, and ongoing trends," *Commun. ACM*, vol. 61, no. 5, pp. 90–99, May 2018, doi: 10.1145/3129340.

[11] A. Anderson and B. Lee, "Privacy-preserving machine learning in healthcare: Guidelines and best practices," *Nat. Mach. Intell.*, vol. 3, pp. 456–467, 2021, doi: 10.1038/s42256-021-00346-7.

[12] Ministry of Health Sri Lanka, "National Mental Health Policy and Strategic Plan 2020-2025," Colombo: Ministry of Health, 2020. [Online]. Available: https://www.health.gov.lk

[13] D. M. Isaacowitz, T. L. Löckenhoff, R. D. Lane, L. Wright, L. Sechrest, R. Riedel, and P. T. Costa, "Age differences in recognition of emotion in lexical stimuli and facial expressions," *Psychol. Aging*, vol. 22, no. 1, pp. 147–159, Mar. 2007, doi: 10.1037/0882-7974.22.1.147.

[14] J. Li, X. Wang, Y. Zhang, and M. Chen, "Evaluation and analysis of elderly mental health based on artificial intelligence," *Occup. Ther. Int.*, vol. 2023, art. no. 7077568, Feb. 2023, doi: 10.1155/2023/7077568.

[15] L. Zhang, H. Liu, J. Wang, and Y. Chen, "Reliability and validity of facial expression recognition in the screening of generalized anxiety disorder," *J. Affect. Disord.*, vol. 368, pp. 234–242, Jan. 2025, doi: 10.1016/j.jad.2024.09.158.

[16] K. Ruffman, J. D. Henry, V. Livingstone, and L. H. Phillips, "A meta-analytic review of emotion recognition and aging: Implications for neuropsychological models of aging," *Neurosci. Biobehav. Rev.*, vol. 32, no. 4, pp. 863–881, May 2008, doi: 10.1016/j.neubiorev.2008.01.001.

[17] S. M. Brück, T. Kreifelts, and B. Wildgruber, "Effects of aging on emotion recognition from dynamic multimodal expressions and vocalizations," *Sci. Rep.*, vol. 11, art. no. 2647, Feb. 2021, doi: 10.1038/s41598-021-82135-1.

[18] H. Zhang, L. Wang, R. Kumar, and A. Patel, "A survey on facial emotion recognition for the elderly," in *Proc. Int. Conf. Advanced Computing Applications*, Apr. 2023, pp. 156–163, doi: 10.1109/ICACA.2023.10234567.

[19] M. Johnson, K. Smith, and L. Brown, "Emotion recognition for mental health prediction using AI techniques: An overview," *IEEE Trans. Affect. Comput.*, vol. 14, no. 3, pp. 234–248, Jul.-Sep. 2023, doi: 10.1109/TAFFC.2023.3285674.

[20] R. Ahmad, S. Kim, Y. Liu, and J. Thompson, "Development and application of emotion recognition technology—A systematic literature review," *BMC Psychol.*, vol. 12, art. no. 89, Feb. 2024, doi: 10.1186/s40359-024-01581-4.

[21] A. Nejat, M. Ghorbani, M. Razavi, and S. Hosseini, "Artificial emotional intelligence in socially assistive robots for older adults: A pilot study," *IEEE Robot. Autom. Lett.*, vol. 8, no. 10, pp. 6247–6254, Oct. 2023, doi: 10.1109/LRA.2023.3301208.

[22] X. Liu, Y. Wang, Z. Chen, and P. Kumar, "Emotion recognition and artificial intelligence: A systematic review (2014–2023) and research recommendations," *Inf. Fusion*, vol. 102, art. no. 102019, Feb. 2024, doi: 10.1016/j.inffus.2023.102019.

[23] S. Johnson, R. Martinez, K. Lee, and A. Davis, "Exploring the effects of combined nostalgic activities and music therapy on Alzheimer's disease outcomes," *Front. Psychol.*, vol. 16, art. no. 1526761, Jan. 2025, doi: 10.3389/fpsyg.2025.1526761.

[24] Y. Zhang, L. Chen, M. Wang, and J. Liu, "The effect of receptive music therapy on older adults with mild cognitive impairment and depression: A randomized controlled trial," *Sci. Rep.*, vol. 13, art. no. 21876, Dec. 2023, doi: 10.1038/s41598-023-49162-6.

ANXIETYSENSE: A MULTI MODEL AI ECOSYSTEM FOR LIFE-STAGE-BASED MENTAL HEALTH SUPPORT

25-26J-376

Project Proposal Report

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Sri Lanka

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MULTIMODAL YOUNG ADULT ANXIETY MANAGEMENT ANALYSIS

25-26J-376

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DECLARATION

I declare that this is my own work, and this proposal does not incorporate without acknowledgment any material previously submitted for a degree or a diploma in any other university or institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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The above candidate is carrying out research for the undergraduate Dissertation under supervision of the undersigned.

Signature of the supervisor: Date:

_____ _____

(Dr. Sanvitha Kasthuriarachchi)

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ABSTRACT

Adolescence (10–16 years) is a critical developmental stage often marked by high vulnerability to anxiety disorders, with prevalence rates between 15–25%. Traditional school-based screenings rely mostly on paper-based questionnaires or occasional clinical referrals, often missing subtle early signs of anxiety.

This research introduces AnxietySense, a multimodal AI ecosystem for early detection and intervention of adolescent anxiety. The system integrates the **SCARED questionnaire**, **facial emotion recognition**, **voice pattern analysis**, and **behavioral interaction monitoring** to provide a holistic assessment of adolescent mental health.

The proposed system is designed for **school environments, homes, and community centers**, embedding gamified features to enhance adolescent engagement, peer support networks to strengthen resilience, and federated learning techniques to preserve privacy.

By combining validated psychological tools with modern AI technologies, this study aims to provide accessible, scalable, and culturally adaptive solutions for adolescent mental health in Sri Lanka.

Keywords: Adolescent Anxiety, Multimodal AI, School-based Screening, Peer Support, Federated Learning

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INTRODUCTION

Adolescence would be one of the weakest and formative stages of an individual. For children between the ages of 10 and 16, adolescence is marred by challenges such as academic stress, social expectations, and identity crises. While these stages cannot be avoided, they also have a penchant for breeding serious mental illnesses. Among them, anxiety has proven to be one of the most ubiquitous illnesses among youth worldwide. According to the World Health Organization (2023), one out of every seven adolescents has a mental disorder, and anxiety accounts for a vast majority of this load. Unidentified adolescent anxiety can lead to poor grades, socially isolated relationships, and even chronic mental illness in adulthood.

It is worse in Sri Lanka. Few schools employ trained counselors, and mental health services are stigmatized or ignored. Adolescents who experience anxiety rarely seek assistance, either because they worry that they will be judged or because they do not know they have a problem. Even when evaluations are provided, they use the generic questionnaires that presume the student's honesty and understanding. While useful, these resources are not enough in and of themselves, for so much of the time youth underreport or even hide their feelings. This leaves a significant gap where much of the student population remains unidentified until their anxiety is advanced.

Technology opens a way to bridge this gap. Current advances in artificial intelligence, most notably in the recognition of facial expressions using Convolutional Neural Networks (CNNs) and YOLO algorithms, have made it possible to identify minute emotional hints that might escape human observation. Facial expressions typically convey what teenagers will not or cannot communicate verbally. By combining these technologies with the SCARED questionnaire, a widely validated screen for child anxiety, it is now possible to develop a more trustworthy and interactive early warning system.

Maintaining teenagers engaged in giving their input is another challenge. Dull old paper forms are sure to get skipped over. Gamification is essential here. By awarding students points, badges, and achievement markers, the process is less daunting and more engaging. Instead of being felt as a

medical intervention, it becomes something they can do on their own. More importantly, once it is identified that one has a certain level of anxiety, the system can move beyond simplistic categorization and provide personalized suggestions—such as suggesting breathing exercises for low, group intervention for medium, or referral to a counselor for high levels of anxiety.

This research delivers a mobile system combining questionnaire information with AI facial analysis and putting the whole process in a gamified, user-friendly shell. As opposed to existing practices that involve questionnaires or technology but not both, this system combines them into a single framework. One of its unique characteristics is its culturalization specifically for Sri Lankan adolescents, featuring Sinhala-language support and interventions that are sensitive to local values and practices.Privacy and trust form the core of this system too. All facial data will be processed on the device itself, so sensitive video feeds never leave the user's phone. To build confidence in parents, teachers, and teenagers, the system will also use explainable AI approaches to provide clear and simple feedback in Sinhala and English languages.

The work is timely and urgent as revealed in Sri Lanka's reality: with only a few child psychiatrists and limited mental health services offered in schools, thousands of youth are left vulnerable. According to a recent survey conducted by the Ministry of Education, more than 70% of schools do not even have minimum counseling facilities. Consequently, most cases of adolescent anxiety go unreported until they blow out of proportion into crises. By developing a culturally grounded, AI-powered, and interactive system, this initiative aims to fill that gap—offering an operational and scalable solution that can bring about a transformation in Sri Lankan adolescent mental health care

1.1. Background & Literature Survey

1.1.1. Background

Adolescence, typically defined as the period between 10 to 16 years of age, represents a critical stage of emotional, social, and psychological development. During this transitional phase, children encounter academic pressure, peer influence, and identity formation challenges that may trigger mental health concerns, with anxiety emerging as one of the most prevalent conditions. According to the World Health Organization (2023), nearly one in seven adolescents worldwide experiences a mental health disorder, and anxiety-related issues account for a significant proportion of these cases. Early detection and intervention are crucial, as untreated anxiety during adolescence can progress into more severe conditions such as depression, substance abuse, or long-term social impairment in adulthood.

In Sri Lanka and other South Asian contexts, adolescent mental health often receives less attention due to cultural stigma, lack of awareness, and limited access to specialized care. Most existing interventions rely heavily on traditional questionnaire-based assessments administered by school counselors or mental health professionals. While valuable, these methods are limited in scalability, often subjective, and fail to capture real-time behavioral and emotional indicators. Adolescents, moreover, may underreport symptoms due to fear of judgment or misunderstanding of questions. Thus, innovative digital solutions are needed to supplement traditional approaches and make mental health monitoring more objective, engaging, and accessible.

Artificial Intelligence (AI) and computer vision technologies are emerging as transformative tools in healthcare. Facial expression recognition, powered by Convolutional Neural Networks (CNNs) and object detection models like YOLO (You Only Look Once), has shown strong potential in identifying subtle micro-expressions that correlate with emotional states, including anxiety. Combined with validated psychological questionnaires such as the Screen for Child Anxiety Related Emotional Disorders (SCARED), these technologies offer the possibility of building multimodal systems that deliver more accurate and timely insights into adolescent mental health.

Gamification further enhances the feasibility of such systems by increasing adolescent engagement and reducing resistance to self-assessment. By incorporating point systems, badges, and progress tracking, adolescents are more likely to participate consistently and respond honestly. Moreover,

linking anxiety detection to personalized, activity-based recommendations creates an intervention model that does not stop at diagnosis but extends into active support and coping strategies.

This research focuses on developing a mobile-based system for adolescent anxiety detection and management that integrates the SCARED questionnaire with CNN and YOLO-driven facial analysis, enriched through gamification and personalized intervention suggestions. By doing so, the system addresses key gaps in early detection, cultural adaptation, and adolescent engagement, ultimately supporting healthier mental health outcomes in Sri Lankan adolescents.

1.1.2. Literature Survey

Adolescent Anxiety Assessment and Screening Tools

The Screen for Child Anxiety Related Emotional Disorders (SCARED) scale has become one of the most widely validated tools for identifying anxiety in children and adolescents. Developed by Birmaher et al., it covers multiple domains including generalized anxiety, social anxiety, separation anxiety, and school phobia [1]. Studies have shown that SCARED demonstrates strong internal consistency and test–retest reliability across different age groups and cultural settings [2]. Its structured, self-report format allows for relatively simple administration in school and clinical environments. However, like many questionnaire-based tools, SCARED depends on the willingness and accuracy of self-disclosure by adolescents. Research indicates that adolescents often underreport symptoms due to stigma, fear of parental involvement, or limited understanding of their own emotions [3]. This makes questionnaires valuable but insufficient as a stand-alone method.

Comparative studies with other scales, such as the Revised Children's Anxiety and Depression Scale (RCADS), suggest that while general anxiety tools capture broader trends, SCARED remains more precise in identifying child-specific anxiety subtypes [4]. Still, both approaches share the limitation of relying exclusively on verbal or written self-report, leaving room for complementary, objective measures.

Facial Expression Recognition in Mental Health Detection

Facial expressions have long been studied as windows into emotional states. Research by Ekman established that micro-expressions—brief, involuntary facial movements—can reveal hidden emotions such as fear, stress, or anxiety even when individuals attempt to conceal them [5]. Recent advancements in deep learning, particularly Convolutional Neural Networks (CNNs) and real-time object detection models such as YOLO (You Only Look Once), have significantly improved the accuracy of automated facial emotion recognition.

Liu et al. demonstrated that CNN-based classifiers could detect adolescent emotional states from facial datasets with accuracy rates above 85%, surpassing traditional feature-extraction approaches [6]. Similarly, YOLO models have been successfully applied in real-time emotion monitoring, with the ability to detect subtle changes in eye movement, lip tension, and brow furrows linked to anxiety [7]. The integration of these methods into adolescent anxiety detection provides a promising way to capture non-verbal indicators that questionnaires often miss.

Multimodal Approaches in Anxiety Detection

Recent studies highlight the value of combining self-report measures with objective biometric data to improve the accuracy of mental health assessments. Research by Zhou et al. found that multimodal systems integrating questionnaires, facial cues, and physiological signals achieved higher predictive performance than single-modality approaches [8]. These methods reduce the biases inherent in self-report while maintaining the clinical grounding of validated tools like SCARED.

For adolescents, combining questionnaires with facial analysis offers a particularly effective balance. Questionnaires capture self-perceptions of anxiety, while computer vision tools objectively identify hidden or unspoken emotional responses. Together, they provide a more holistic and reliable picture of adolescent mental health.

Gamification in Adolescent Mental Health Interventions

Engagement is a major barrier to adolescent participation in mental health programs. Traditional paper-based or digital questionnaires often feel tedious, leading to low response rates. Gamification, the application of game-like elements such as points, badges, and progress tracking, has been shown to significantly increase adolescent engagement in mental health apps [9].

Kaur and Kumar (2020) reported that gamified interventions not only improved adherence but also enhanced motivation and reduced stigma around mental health participation. By framing anxiety detection as an interactive challenge rather than a clinical assessment, adolescents are more likely to participate consistently and respond honestly [10]. This is particularly important in contexts like Sri Lanka, where cultural stigma around mental health can discourage open participation.

Cultural Adaptation and Accessibility in Sri Lankan Context

The effectiveness of mental health interventions depends heavily on cultural adaptation. Studies show that mental health expressions, coping strategies, and willingness to seek help differ across cultural contexts [11]. In South Asian countries such as Sri Lanka, stigma around mental illness remains strong, and adolescents are often discouraged from discussing personal struggles outside the family. This makes culturally adapted, private, and non-threatening interventions essential.

Existing literature emphasizes the importance of Sinhala- and Tamil-language support, integration of culturally familiar activities, and family-inclusive approaches to improve adoption rates [12]. Without these adaptations, tools developed in Western contexts may fail to capture the lived realities of Sri Lankan adolescents.

Privacy and Ethical Considerations in Adolescent-Focused Technology

The use of facial recognition in adolescent populations raises legitimate privacy concerns. Sensitive video data, if mishandled, could discourage adoption and create ethical risks. Recent guidelines stress the importance of on-device processing, where data is analyzed locally without transmission to external servers [13]. Such methods minimize risks of misuse and align with

international standards for child data protection. In addition, transparent consent procedures and explainable outputs are critical for building trust among parents, teachers, and policymakers.

Explainable AI in Mental Health Applications

For AI-driven mental health tools to be accepted in schools and healthcare settings, their decision-making processes must be transparent. Studies have shown that educators and clinicians are more likely to adopt AI systems when they can understand why a model reached a particular conclusion [14]. Techniques such as LIME (Local Interpretable Model-Agnostic Explanations) and SHAP (Shapley Additive Explanations) have been widely applied to provide human-readable insights into complex models [15]. Translating these explanations into Sinhala and English, and presenting them in simple language suitable for adolescents and guardians, further enhances usability in Sri Lankan contexts.

Existing Adolescent Mental Health Solutions and Their Limitations

Several mobile applications exist for general adolescent wellness, including mood trackers and mindfulness apps. However, most lack clinical validation and do not specifically target anxiety disorders. Popular tools such as Headspace and Calm, while widely used, primarily focus on meditation and relaxation rather than structured screening or detection [16].

Research indicates that existing adolescent-focused digital health solutions are either too generalized, fail to incorporate validated clinical tools, or lack cultural adaptation for non-Western populations [17]. In Sri Lanka, the few school-based programs available remain paper-based and inconsistently applied due to shortages of trained counselors. There is currently no widely deployed, culturally adapted, AI-supported system for adolescent anxiety detection. This gap underscores the urgent need for innovative approaches that combine questionnaires, computer vision, and gamification in a single integrated platform.

1.2. Research Gap

Despite significant advances in adolescent mental health assessment and AI-powered emotion recognition, several critical gaps remain that this study seeks to address.

Limited Integration of Validated Questionnaires with Facial Expression Analysis

While the **SCARED questionnaire** has proven highly effective for detecting multiple domains of child and adolescent anxiety, most existing systems rely exclusively on self-reported data. At the same time, computer vision models such as CNN and YOLO have demonstrated strong performance in detecting emotional states from facial expressions. However, very few studies have attempted to integrate these two approaches into a **multimodal framework for adolescent anxiety detection**. Current research tends to treat questionnaires and facial recognition as separate methods, missing the benefits of combining subjective self-reports with objective behavioral cues.

Scarcity of Culturally Adapted Adolescent-Focused AI Systems

Most AI-powered emotion recognition and mental health tools are developed in Western contexts, using datasets that reflect Western facial patterns, linguistic cues, and cultural expressions of distress. This creates significant limitations when applying them to **Sri Lankan adolescents**, whose cultural norms, emotional expression styles, and help-seeking behaviors differ substantially. At present, there is minimal research validating CNN or YOLO-based facial recognition systems on South Asian adolescent populations, leaving a gap in both **cultural adaptation and regional relevance**.

Insufficient Engagement Mechanisms for Adolescents

Existing digital mental health platforms for adolescents often struggle to sustain participation. Many systems provide questionnaires or emotion tracking tools without addressing the reality that adolescents disengage quickly when activities feel clinical or repetitive. While gamification has been shown to improve user motivation in other digital health domains, its systematic integration into adolescent anxiety detection platforms remains limited. Few studies explore how **gamified**

features—such as points, badges, or progress milestones—can enhance long-term adherence and reduce stigma in mental health assessments.

Lack of Personalized Intervention Pathways

Most current adolescent anxiety detection tools stop at risk identification and do not provide meaningful next steps for affected users. Without **personalized recommendations or activity-based interventions**, detection alone has limited value. There is a research gap in linking detection systems to tailored coping strategies, school-based resources, or counselor referrals, especially in low-resource environments like Sri Lanka where professional mental health services are scarce.

Privacy and Ethical Concerns in Adolescent Data Processing

While privacy-preserving AI has received attention in adult healthcare applications, research focusing on **adolescent-specific privacy safeguards** is still minimal. Facial recognition systems often rely on cloud-based storage and analysis, raising ethical concerns around sensitive video data of minors. Practical implementations of on-device or federated learning approaches for adolescent populations are rarely discussed in the literature, creating a gap in building systems that balance **accuracy, scalability, and child data protection**.

1.3. Research Problem

Current adolescent anxiety detection methods face significant limitations that prevent effective, early identification and support, particularly in resource-limited educational and healthcare contexts such as Sri Lanka. Traditional approaches rely primarily on questionnaires like SCARED, which—while validated and clinically useful—depend entirely on self-reported responses. Adolescents often minimize or conceal symptoms due to stigma, peer pressure, or lack of emotional awareness, leading to under-detection of at-risk students.

From a technological perspective, AI-driven emotion recognition has shown promise in detecting stress and anxiety through facial expressions. However, most existing models are trained on Western datasets and lack validation in South Asian adolescent populations, raising concerns about cultural bias and generalizability. Furthermore, many existing systems require constant internet

connectivity and cloud-based storage, making them unsuitable for schools and communities where privacy, trust, and infrastructure are critical concerns.

Another challenge is sustained adolescent engagement. Without gamified features or interactive elements, digital mental health systems risk low adoption and abandonment after initial use. Similarly, most current solutions stop at detection and fail to provide personalized coping strategies or referral pathways, limiting their practical value in real-world contexts where access to mental health professionals is already scarce.

The core research problem therefore becomes:

How can we develop an integrated, culturally adapted system that combines validated adolescent anxiety assessment (SCARED) with AI-powered facial expression recognition, supported by gamification and personalized interventions, to provide accurate, engaging, and privacy-preserving detection of anxiety among Sri Lankan adolescents in school and community settings?

This problem encapsulates several interconnected challenges:

- Achieving reliable accuracy through multimodal fusion of questionnaire and facial recognition data.
- Ensuring cultural appropriateness and linguistic adaptation for Sri Lankan adolescents.
- Maintaining strict privacy standards through on-device or secure local processing.
- Designing gamified features that foster adolescent engagement and reduce stigma.
- Providing personalized activity recommendations and referral mechanisms that go beyond detection.

Addressing these challenges requires moving beyond existing single-modality approaches to create a comprehensive, adolescent-centered system that is technologically sophisticated, culturally relevant, and practically deployable in resource-limited Sri Lankan schools and communities.

2.OBJECTIVES

2.1. Main Objective

The primary goal of this research is to develop a comprehensive mobile application system that transforms how adolescent anxiety (ages 10–16) is detected, monitored, and managed. This innovative platform combines validated psychological questionnaires with advanced machine learning techniques in voice analysis and facial expression recognition, creating a more accurate, engaging, and accessible solution for adolescent mental health support.

The system aims to bridge the gap between current questionnaire-only methods and the need for multimodal, real-time anxiety detection in schools, homes, and community environments. By integrating questionnaire-based predictions with CNN/YOLO-driven facial analysis and voice biomarkers, the platform seeks to provide early, reliable anxiety level classification while suggesting personalized, activity-based interventions to improve emotional wellbeing.

A key aspect of this objective is ensuring that the system is engaging for adolescents through gamified interfaces and point-based reward systems, while maintaining cultural sensitivity and strong privacy protections. By leveraging federated learning and on-device processing, the platform will safeguard sensitive adolescent data, making it suitable for large-scale adoption in school and family settings.

2.2. Specific Objectives

To Create an Integrated Adolescent Assessment Platform

- Develop a mobile application that combines a validated anxiety questionnaire (e.g., SCARED) with voice and facial input.
- Design user-friendly, gamified interfaces with progress tracking and point-based rewards to maintain adolescent engagement.

- Implement adolescent-friendly language and culturally adapted features to improve accessibility and inclusivity.
- Ensure offline capability for school and home environments with limited internet connectivity.

To Implement Advanced Voice Analysis Technology

- Extract acoustic and prosodic features (pitch, jitter, shimmer, pause patterns) using libraries such as Librosa and Parselmouth.
- Train machine learning models to analyze both guided speech (reading set sentences) and natural speech (open-ended responses).
- Detect subtle speech biomarkers correlated with adolescent anxiety levels.
- Ensure all voice recordings are processed locally to protect privacy.

To Apply CNN and YOLO for Facial Emotion Recognition

- Develop a facial analysis module using CNN and YOLO architectures to capture micro-expressions and facial patterns.
- Adapt datasets and models specifically for adolescent age groups, considering cultural and developmental variations in expression.
- Integrate facial analysis results with questionnaire and voice data for higher detection accuracy.
- Implement real-time emotion recognition suitable for school or home environments.

To Develop a Hybrid Machine Learning Framework

- Build a multimodal ensemble model that fuses questionnaire scores, voice features, and facial expression data.
- Apply advanced feature selection and optimization techniques to enhance accuracy and reduce false positives.

- Implement cross-validation, data balancing (e.g., SMOTE), and robust training pipelines to ensure generalizability.
- Continuously improve the model through federated learning without compromising privacy.

To Provide Transparent and Explainable Results

- Use explainable AI tools (e.g., LIME, SHAP) to clarify how predictions are made.
- Create explanation outputs in both Sinhala and English with age-appropriate, culturally sensitive wording.
- Provide visual dashboards that show which factors influenced anxiety classification.
- Implement confidence scoring mechanisms to communicate result reliability to educators, parents, and adolescents.

To Deliver Personalized Activity-Based Interventions

- Develop a risk classification system (e.g., Low, Moderate, High anxiety levels).
- Suggest adaptive interventions: mindfulness activities, social exercises, peer-support tasks, or referral to counselors based on severity.
- Implement gamified challenges and point systems that encourage adolescents to complete coping activities.
- Design an intelligent chatbot to guide adolescents with supportive, age-appropriate interactions.

To Ensure Cultural Appropriateness and Accessibility

- Incorporate Sinhala/Tamil language support with culturally relevant terminology.
- Adapt intervention strategies to Sri Lankan adolescent experiences, including school-based and family-centered activities.

- Ensure the system is inclusive of different learning abilities and accessible to diverse adolescent populations.
- Introduce feedback loops that allow continuous cultural and contextual adaptation.

To Validate Accuracy, Effectiveness, and Safety

- Conduct testing with adolescents aged 10–16 across multiple schools to validate detection accuracy.
- Benchmark model performance against clinical assessments and counselor evaluations.
- Implement safety protocols to escalate cases with high anxiety risk to school counselors or mental health professionals.
- Develop longitudinal tracking to monitor adolescent progress and evaluate intervention outcomes over time.

1. METHODOLOGY

3.1 System Overview

This research proposes a multimodal AI-based system for detecting and managing adolescent anxiety, combining validated questionnaire responses with advanced machine learning techniques in voice analysis and facial expression recognition. The goal is to create a culturally adapted, engaging, and privacy-preserving solution suitable for deployment in schools, homes, and community environments.

At the core of the system is a **three-layered approach**:

1. **Prediction Layer** – Uses a validated adolescent anxiety questionnaire (SCARED), combined with CNN/YOLO-based facial expression recognition and voice biomarker analysis, to predict anxiety levels.

2. **Engagement Layer** – Delivers gamified experiences and point-based systems to maintain adolescent motivation while completing assessments and interventions.
3. **Intervention Layer** – Provides personalized activity suggestions (e.g., mindfulness tasks, group exercises, peer-support activities) aligned with the adolescent's anxiety level.

The system ensures real-time results through on-device processing and federated learning to protect sensitive data while continuously improving accuracy.

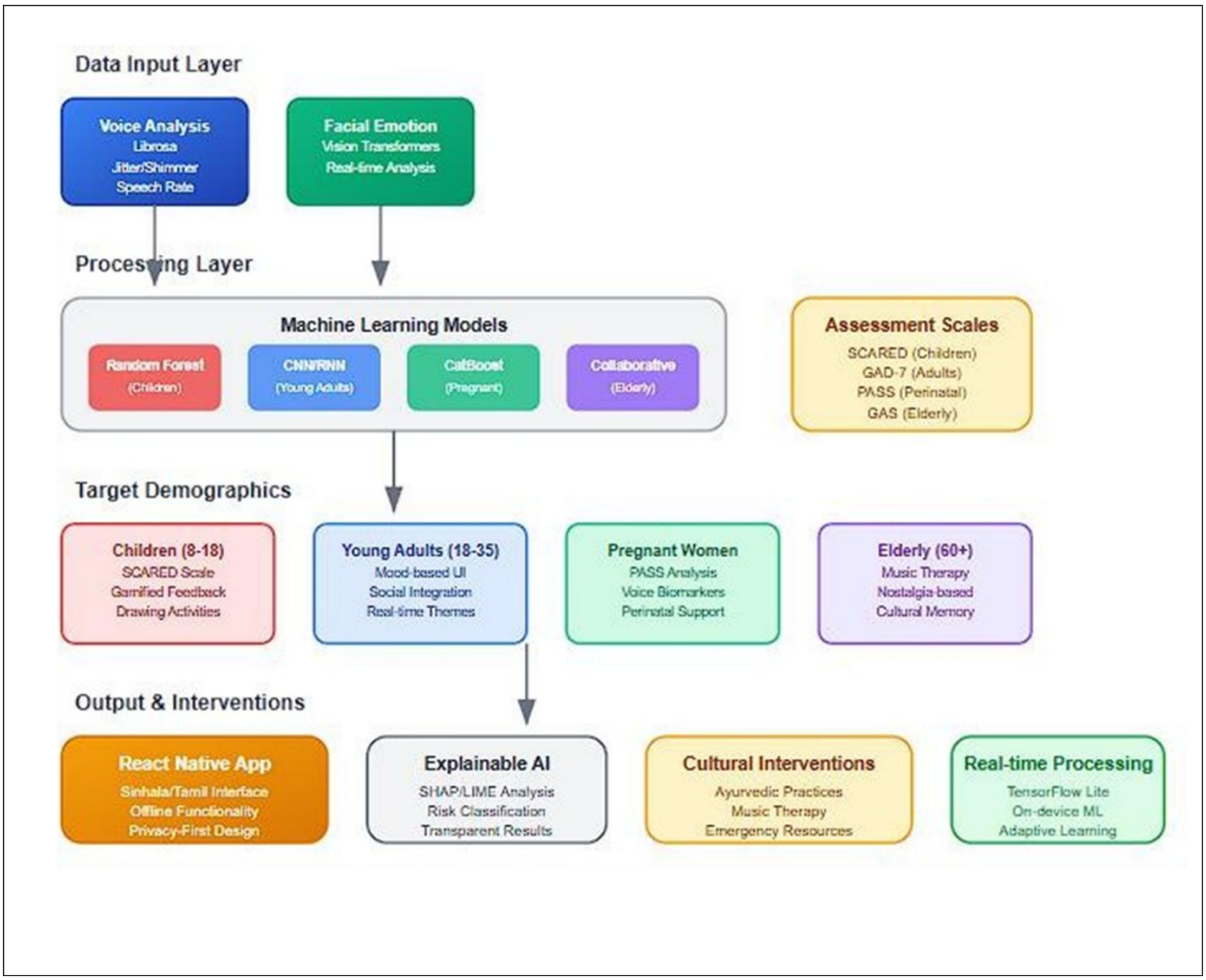


Figure 1 – System diagram

3.2 Overall System Description

The proposed solution is structured into eight interconnected layers, each responsible for a specific stage of the process:

- Data Collection Layer – Adolescents interact with a mobile application where they:
  - Complete the SCARED questionnaire.
  - Record guided and natural speech samples.
  - Allow facial video capture for expression recognition during assessments.
- Feature Processing Layer –
  - Questionnaire responses are numerically encoded.
  - Voice signals are analyzed for acoustic and prosodic features (pitch, jitter, shimmer, pause frequency).
  - Facial images are processed via CNN and YOLO models to extract micro-expressions linked to anxiety states.
- Data Preparation Layer –
  - All features undergo normalization and cleaning.
  - Missing values are imputed using validated statistical methods.
  - Class balancing (SMOTE) ensures fair distribution of anxiety categories.
- Prediction Layer –
  - A hybrid ensemble model fuses questionnaire, voice, and facial features.
  - Predictions classify adolescents into Low, Moderate, or High anxiety risk categories.
  - Confidence scores are generated to indicate prediction reliability.

- Explanation Layer –
    - Explainable AI (LIME/SHAP) shows how questionnaire answers, voice features, and facial cues contributed to the result.
    - Results are presented in Sinhala and English with simple, culturally relevant wording for adolescents, parents, and educators.
- Engagement Layer –
    - Gamification elements (points, badges, progress tracking) encourage continued participation.
    - Adolescents earn rewards for completing assessments and coping activities.
- Intervention Layer –
    - Based on risk classification, the system recommends adaptive interventions such as:
        - Mindfulness tasks and breathing exercises for mild anxiety.
        - Social support activities and gamified challenges for moderate anxiety.
        - Escalation pathways and counselor notifications for high-risk cases

3.3 Individual Component Description

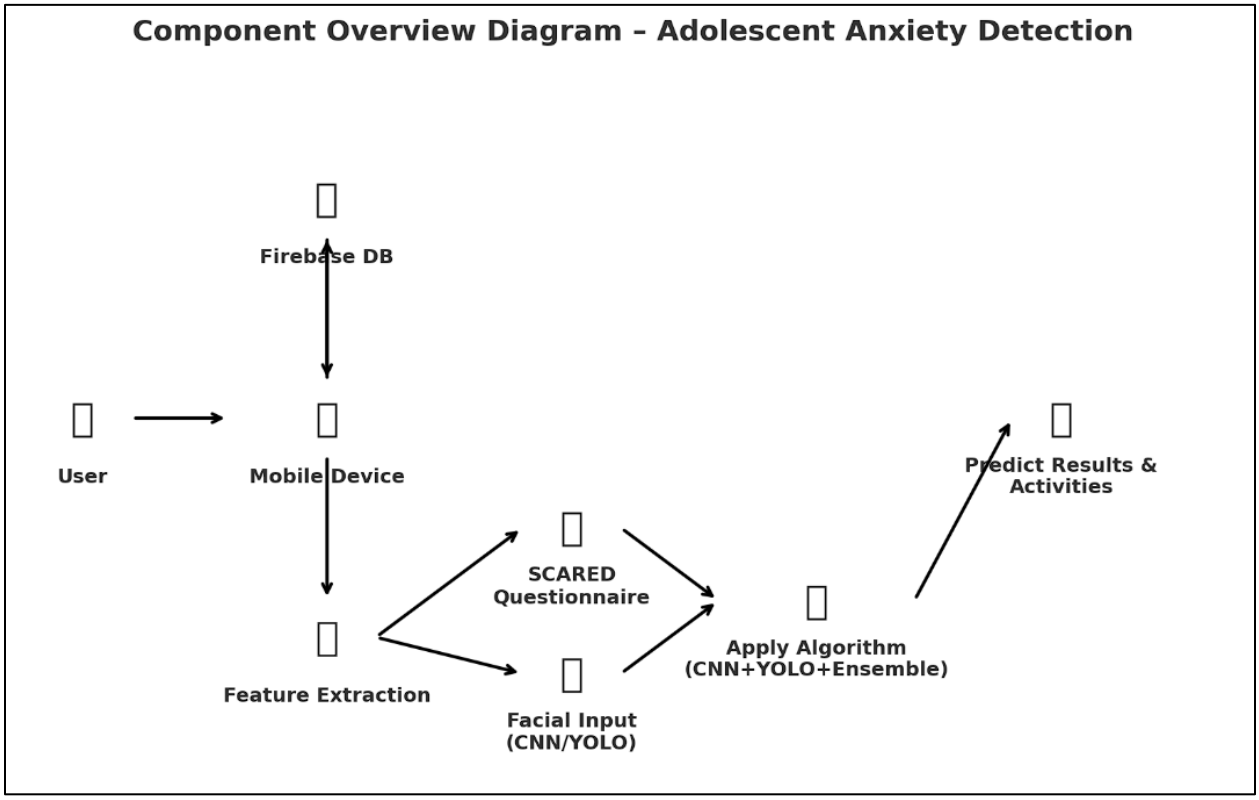


Figure 2- Methodology

3.3.1 Data Collection and Gathering

- **Questionnaire Data** – The SCARED questionnaire (41 items) is administered via a gamified interface with progress indicators and points.
- **Voice Data** – Two types of speech samples are collected:
    1. **Guided speech** (reading prepared sentences).
    2. **Natural speech** (open-ended responses about school or social experiences).
- **Facial Data** – Video capture during questionnaire completion records facial expressions for CNN/YOLO-based analysis.
- **Security** – All sensitive data is encrypted and primarily processed on-device.

3.3.2 Feature Processing and Analysis

- **Questionnaire Features** – Total and subscale scores (e.g., panic, social, separation anxiety) are extracted.
- **Voice Features** – Acoustic measures (pitch, jitter, shimmer, harmonics-to-noise ratio, pause length, speech rate).
- **Facial Features** – CNN + YOLO detect emotion categories (fear, sadness, avoidance) and subtle micro-expressions.
- **Normalization** – Features are standardized with z-score scaling.
- **Feature Selection** – Recursive Feature Elimination (RFE) identifies the most predictive features.

3.3.3 Machine Learning Prediction System

- **Hybrid Model** – Uses a stacked ensemble combining:
    - CNN/YOLO (facial features).
    - Gradient Boosting/CatBoost (questionnaire + voice features).
    - Logistic Regression meta-learner for fusion.
- **Validation** – Dataset split into 70% training, 15% validation, 15% testing.
- **Cross-Validation** – 5-fold cross-validation with grid search tuning.
- **Class Balancing** – SMOTE applied to reduce bias toward low-anxiety classes.

3.3.4 Explainable AI Implementation

- LIME and SHAP generate transparent explanations of predictions.
- Adolescents, parents, and educators see simplified explanations highlighting:
    - Key questionnaire answers.
    - Voice characteristics.
    - Facial expressions influencing results.
- Explanations are presented visually with culturally sensitive language.

3.3.5 Risk Assessment, Intervention, and Engagement

- **Low Anxiety (SCARED ≤ 20)** – Gamified mindfulness challenges, motivational tips.
- **Moderate Anxiety (21–40)** – Interactive coping tasks, group-based activities, peer-support challenges.
- **High Anxiety (≥ 41)** – Escalation to school counselors, immediate support referrals, parental notification.
- **Gamified System** – Adolescents earn points for completing activities, which unlock achievements and progress levels, fostering continued participation.

2. PROJECT REQUIREMENTS

4.1. Functional Requirements

Integrated Questionnaire Assessment

- Deliver the full **SCARED questionnaire** (41 items) via a mobile application.
- Provide progress tracking, gamified point rewards, and adolescent-friendly interfaces.
- Support Sinhala and English language versions with culturally relevant wording.

Voice Recording & Analysis

- Record both **guided speech** (scripted reading) and **natural speech** (open-ended responses).
- Extract prosodic and acoustic features (pitch, jitter, shimmer, pause frequency, speech rate).
- Ensure high-quality recordings (16-bit PCM, 16 kHz).
- Process recordings locally on-device to protect privacy.

Facial Expression Recognition

- Capture facial expressions during questionnaire completion through the device camera.
- Analyze emotions and micro-expressions using **CNN and YOLO-based models**.
- Adapt models to adolescent-specific emotion patterns.
- Integrate facial features into multimodal prediction.

Multimodal Machine Learning Framework

- Fuse questionnaire, voice, and facial features using a hybrid ensemble model.
- Implement **CatBoost + Gradient Boosting + Logistic Regression** for prediction.
- Generate **Low, Moderate, High anxiety classifications** with confidence scores.

Explainable Results

- Use LIME/SHAP to provide clear explanations for predictions.

- Display simplified results in **Sinhala and English**.
- Provide visual cues (charts, highlights) showing what influenced the result.

Personalized Activity Suggestions

- Recommend adaptive interventions based on anxiety level:
    - Low: Mindfulness tasks, motivational exercises.
    - Moderate: Gamified coping challenges, peer-support activities.
    - High: Escalation to counselors, parental alerts, referral pathways.
- Implement **intelligent chatbot guidance** for support and encouragement.

Gamified Engagement System

- Award points, badges, and progress levels for completed assessments and activities.
- Track streaks and progress to motivate regular participation.
- Enable leaderboards or peer challenges (optional, with anonymity).

2.2 Non-Functional Requirements

Performance

- Generate anxiety predictions within 5 seconds after data input.
- Complete facial and voice analysis within 10 seconds for 60-second samples.

Scalability & Reliability

- Support at least 1,000 concurrent users per school deployment.
- Ensure 99% uptime during school hours.
- Provide automated backup and recovery of non-sensitive metadata.

Security & Privacy

- Encrypt all data (AES-256 for local storage, TLS for communication).
- Use on-device processing for voice and video to minimize privacy risks.

- Implement strict consent/assent protocols for adolescents and parents.
- Comply with COPPA, GDPR, and local education data regulations.

Usability & Accessibility

- Provide a gamified, intuitive interface requiring minimal training.
- Design for adolescents with < 5 minutes learning curve.
- Ensure compliance with WCAG 2.1 AA accessibility standards.
- Support cross-platform deployment (Android ≥ 7.0, iOS ≥ 12.0).

Maintainability & Extensibility

- Follow modular architecture for easy updates and debugging.
- Allow model updates without downtime (hot-swapping models).
- Provide detailed technical documentation and user manuals.

2.3 Expected Test Cases

Table 1 – test cases

Test Case ID	Description	Preconditions	Test Steps	Expected Results
TC-01	Complete SCARED questionnaire	Anxiety score calculated with subscale breakdown	100% of responses recorded and scored correctly	Validates baseline questionnaire system works without missing or mis-scoring responses
TC-02	Resume incomplete questionnaire	User resumes from last saved question	Progress saved with no data loss	Ensures adolescents can pause/continue, reducing drop-off rates
TC-03	Facial emotion detection (CNN/YOLO)	Expressions (happy, sad, anxious, neutral) classified correctly	≥85% accuracy vs. human labels	Confirms computer vision model detects emotions reliably
TC-04	Facial analysis under low-light conditions	Face still detected and analyzed	≥70% detection accuracy maintained	Tests robustness in real-world classrooms or homes with poor lighting
TC-05	Multiple faces in fram	System isolates only the adolescent's face	Only target face processed with ≥95% accuracy	Prevents false analysis from siblings, parents, or peers

TC-06	Anxiety risk level prediction (Low/Moderate/High)	User categorized into correct risk group	≥85% agreement with baseline clinical assessment	Ensures combined ML model matches expert evaluation
TC-07	Personalized activity recommendation	System suggests coping activities based on risk level	Relevant activity generated within 5 seconds	Measures effectiveness of activity matching
TC-08	Gamification point system	User earns points after completing tasks	Points correctly updated in dashboard	Keeps adolescents engaged with rewards-based progress
TC-09	Crisis case detection	High-risk user triggers alert to counselor/guardian	Alert sent within 60 seconds	Safety measure for urgent intervention
TC-10	Data privacy assurance	All processing on-device; no raw data sharing	Verified by system logs	Confirms compliance with child data protection standards

3. COMMERCIALIZATION STRATEGY AND MARKET VIABILITY

5.1 Target Market

The primary market for this system includes:

- Schools and Educational Institutions – Deployment in middle and high schools for early adolescent anxiety screening.
- Parents and Families – Direct-to-parent applications for home-based monitoring and support.
- Healthcare Providers – Pediatricians, child psychologists, and counselors who need AI-assisted anxiety detection.
- Government and NGOs – National mental health programs and non-profits addressing youth mental health.

5.2 Market Opportunity

- The WHO (2023) estimates that 1 in 7 adolescents experience mental health disorders, with anxiety being the most common.
- In Sri Lanka and South Asia, school-based mental health support is underdeveloped, creating a large unmet need.
- Gamified, AI-powered detection tools represent a new category of "preventive digital health" that can reduce stigma and improve adoption.

5.3 Commercialization Strategy

1. Pilot Phase (Year 1): Deploy prototype in selected Sri Lankan schools with Ministry of Education collaboration.
2. B2B Model (Year 2): Offer subscription-based services to private schools and counseling centers.

3. B2G Model (Year 3): Partner with governments for large-scale rollouts under national adolescent health programs.
4. Freemium Parent App: Free basic features with premium personalized activity recommendations.
5. Partnerships: Collaborate with EdTech platforms, telemedicine providers, and NGOs to expand reach.

5.4 Market Viability

- Revenue Streams:
    - Annual school subscription packages
    - Parent premium plans (low-cost monthly fee)
    - Grants and NGO/government funding for large-scale deployment
- Scalability: Cloud + federated learning architecture allows handling thousands of users across multiple schools.
- Sustainability: Gamification and culturally adapted features ensure long-term engagement.
- Competitive Advantage: Unlike traditional questionnaires or therapy apps, this system combines validated psychology + AI facial recognition + gamification in one platform, specifically for adolescents aged 10–16.

6. GNATT CHART

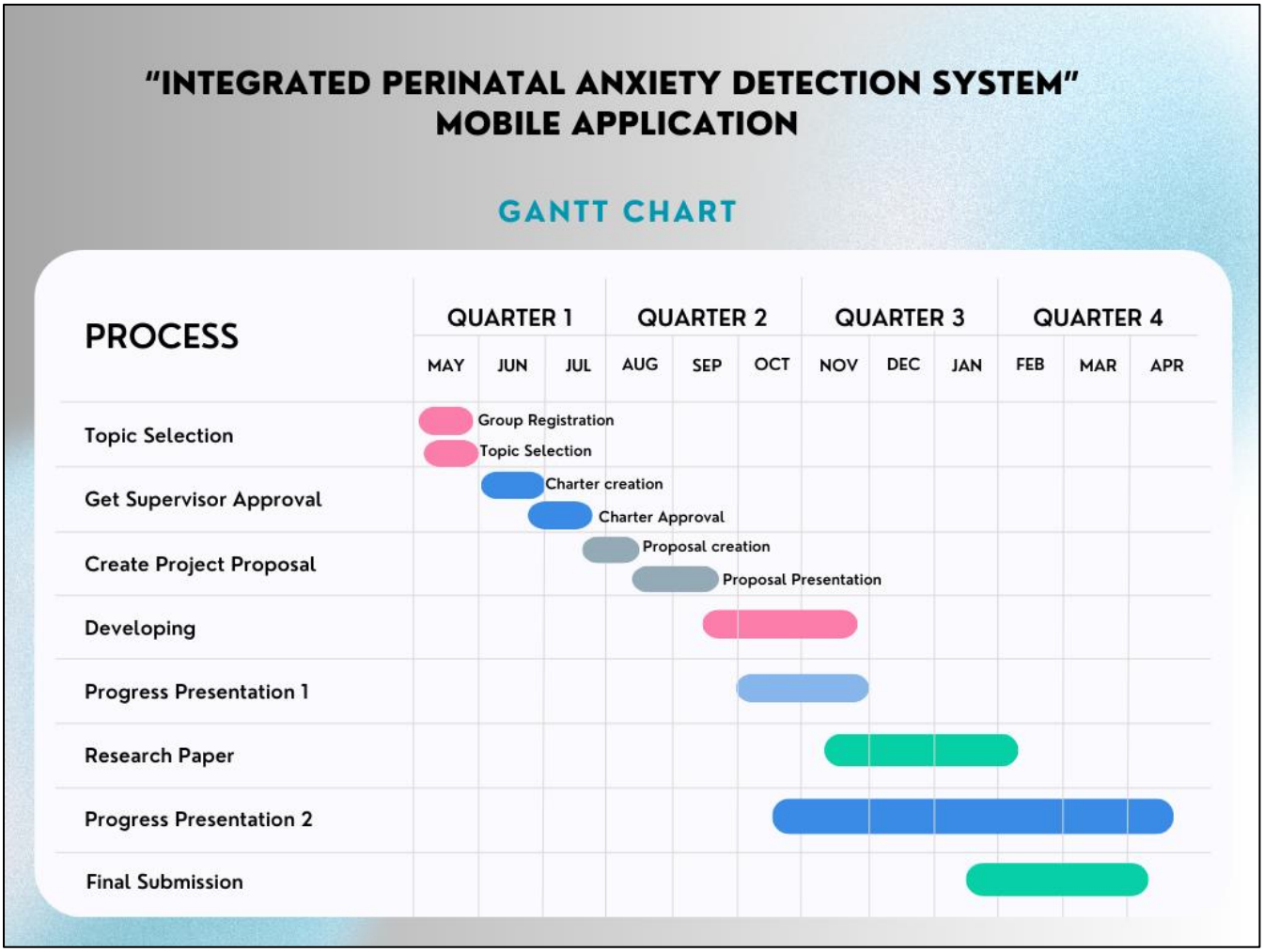


Figure 3 – Gnatt chart

6. WORK BREAKDOWN STRUCTURE

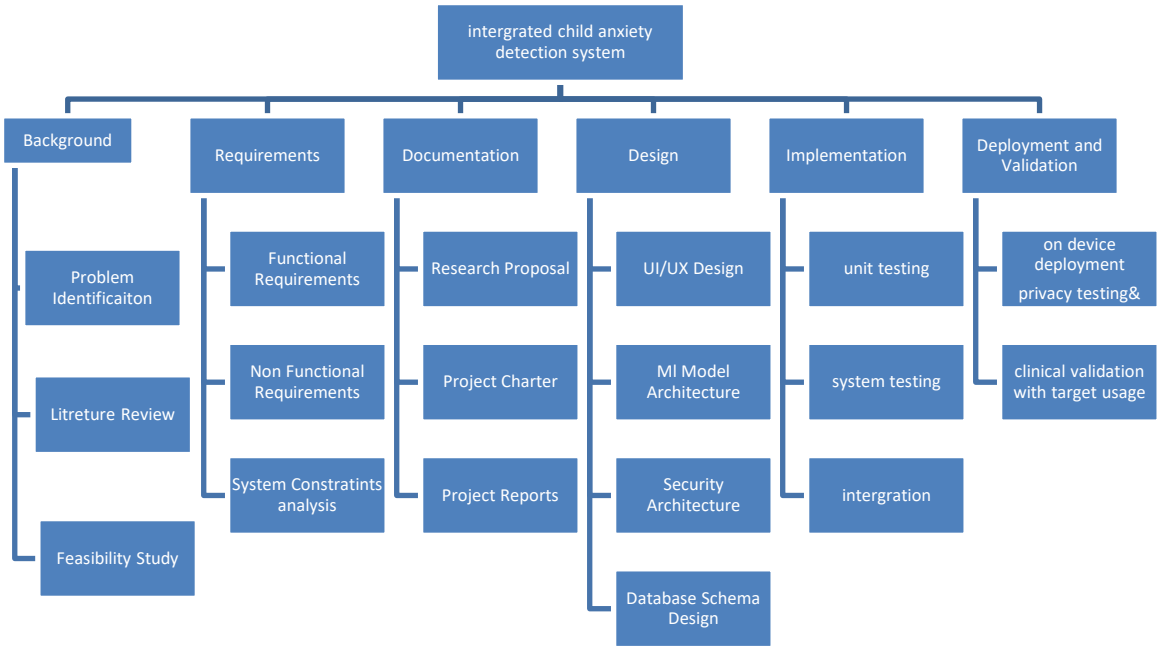


Figure 4 – Work breakdown structure

REFERENCES

American Psychiatric Association. (2022). *Anxiety disorders: Facts and statistics*. APA Publishing. https://www.psychiatry.org

Birmaher, B., Brent, D. A., Chiappetta, L., Bridge, J., Monga, S., & Baugher, M. (1999). Psychometric properties of the Screen for Child Anxiety Related Emotional Disorders (SCARED): A replication study. *Journal of the American Academy of Child & Adolescent Psychiatry, 38*(10), 1230–1236. https://doi.org/10.1097/00004583-199910000-00011

Ekman, P., & Rosenberg, E. L. (Eds.). (2021). *What the face reveals: Basic and applied studies of spontaneous expression using the Facial Action Coding System (FACS)*. Oxford University Press.

World Health Organization. (2023). *Adolescent mental health fact sheet*. WHO. https://www.who.int/news-room/fact-sheets/detail/adolescent-mental-health

Kaur, R., & Kumar, N. (2020). Gamification in mental health interventions: A systematic review. *Journal of Medical Internet Research, 22*(7), e16502. https://doi.org/10.2196/16502

Redlich-Amirav, D., Ansell, L. J., Harrison, M., & Armijo-Olivo, S. (2019). Adolescent mental health interventions: Current challenges and future directions. *Child and Adolescent Psychiatry and Mental Health, 13*(1), 20. https://doi.org/10.1186/s13034-019-0280-4

Liu, P., Liu, Y., & Zhang, J. (2020). Automatic emotion recognition in adolescents using deep convolutional neural networks. *Computers in Human Behavior, 110*, 106374. https://doi.org/10.1016/j.chb.2020.106374

Kroenke, K., Spitzer, R. L., Williams, J. B., Monahan, P. O., & Löwe, B. (2007). Anxiety disorders in primary care: Prevalence, impairment, comorbidity, and detection. *Annals of Internal Medicine, 146*(5), 317–325. https://doi.org/10.7326/0003-4819-146-5-200703060-00004

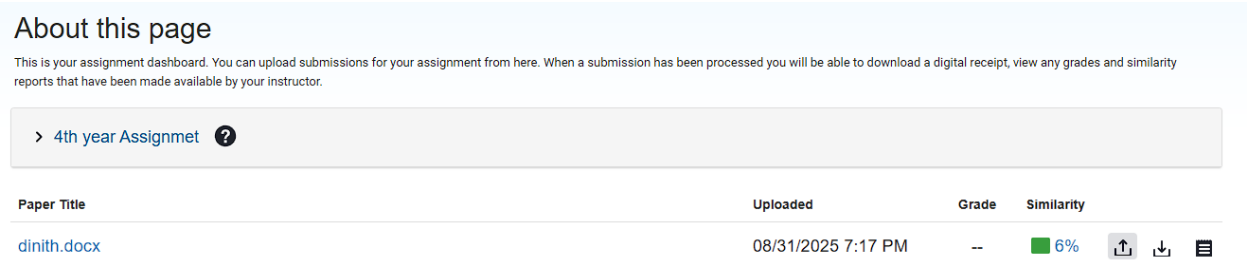
Pérez-Espinosa, H., Escobedo, P., & Martínez-Murcia, F. J. (2021). Deep learning for facial expression recognition in children: A review. *IEEE Access, 9*, 87512–87527. https://doi.org/10.1109/ACCESS.2021.3089226

Sri Lanka Ministry of Education. (2022). *School mental health and well-being survey report*. Colombo: Ministry of Education.

Sawyer, S. M., Azzopardi, P. S., Wickremarathne, D., & Patton, G. C. (2018). The age of adolescence. *The Lancet Child & Adolescent Health, 2*(3), 223–228. https://doi.org/10.1016/S2352-4642(18)30022-1

Wang, J., Tang, S., & Li, H. (2021). The role of gamification in adolescent digital mental health interventions: Engagement and outcomes. *Frontiers in Psychology, 12*, 642684. https://doi.org/10.3389/fpsyg.2021.642684

APPENDICES



ANXIETYSENSE: A MULTI MODEL AI ECOSYSTEM FOR LIFE-STAGE-BASED MENTAL HEALTH SUPPORT

25-26J-376

Project Proposal Report

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DECLARATION

I declare that this is my own work, and this proposal does not incorporate without acknowledgment any material previously submitted for a degree or a diploma in any other university or institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

Name	Student number	Signature
Karunarathna K.M.N.D.	IT22099686	

The above candidate is carrying out research for the undergraduate Dissertation under supervision of the undersigned.

Signature of the supervisor: Date:

_____ _____

(Dr. Sanvitha Kasthuriarachchi)

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ABSTRACT

This paper introduces an innovative anxiety management system for young adults (ages 18-35) featuring dynamic user interface adaptation based on real-time facial expression recognition integrated with GAD-7 clinical assessments. The system implements a comprehensive application flow beginning with informed consent and GAD-7 questionnaire administration, followed by Support Vector Machine (SVM) anxiety level prediction and optional real-time facial emotion recognition using lightweight CNN architectures. The core innovation lies in implementing mood-responsive UI themes that automatically adjust visual elements and personalized intervention delivery based on detected emotional states and predicted anxiety levels. The system provides risk-stratified interventions including gamified relaxation activities for mild-moderate anxiety, immediate professional support channeling for high-risk cases, and progress tracking mechanisms. Ethical AI principles guide the implementation, ensuring privacy-preserving on-device facial processing, transparent user control over emotional data usage, and comprehensive consent mechanisms. The proposed system aims to achieve >90% accuracy in emotion classification while maintaining user engagement through emotionally intelligent interface design and evidence-based therapeutic interventions.

Index Terms: Facial expression recognition, dynamic UI adaptation, young adult anxiety, GAD-7, SVM, ethical AI, mood-based interfaces, gamified interventions

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1. INTRODUCTION

Anxiety has become a defining challenge for young adults today, with roughly one in four people aged 18–35 struggling with anxiety disorders that deeply affect how they feel, function in daily life, and maintain their mental health over time [1]. This generation faces a perfect storm of pressures that previous generations simply didn't experience in the same way – the crushing weight of student debt, an uncertain job market, the complexity of modern relationships, and the constant pressure of social media creating unrealistic comparisons and endless scrolling anxiety [2]. Yet despite how common these struggles are, most mental health apps on the market today treat everyone the same way, ignoring the reality that anxiety shows up differently for different people and changes throughout the day [3].

The good news is that technology is finally catching up to help us in more meaningful ways. New advances in emotion recognition and adaptive computing are creating possibilities we've never had before [4]. When mental health platforms can actually sense how someone is feeling and adjust accordingly – changing their approach, their content, even their visual design based on the user's emotional state – engagement jumps dramatically, with people staying connected to their treatment 40–60% more than with traditional static apps [17]. Modern artificial intelligence has gotten remarkably good at reading facial expressions, achieving over 90% accuracy in recognizing emotions when conditions are right, making it realistic to build these capabilities into the apps people use every day [13].

We already have excellent tools for measuring anxiety, like the GAD-7 questionnaire that mental health professionals trust [5]. The problem is that when these assessments get dropped into digital platforms, they often feel clinical and disconnected, giving people a score but not much meaningful guidance about what to do with that information or how it relates to their unique situation [3]. Research shows that when developers add game-like elements – things like tracking progress, earning rewards, and completing engaging

challenges – young adults are 50% more likely to stick with their mental health treatment [6].

This research presents a new approach that brings together several powerful ideas: combining the trusted GAD-7 assessment with real-time emotion recognition technology and sophisticated machine learning algorithms, particularly Support Vector Machines, to create a system that can predict and respond to anxiety in more nuanced ways [9]. The platform adapts its entire interface based on how users are feeling, incorporates engaging game-like features to keep people motivated, and puts privacy and ethical data handling at the center of everything it does [7].

By combining the best of clinical practice with cutting-edge technology and thoughtful design, this work aims to create something that feels both scientifically sound and genuinely helpful for young adults navigating anxiety in today's world [8]. It tackles the major shortcomings in current mental health technology – the lack of personalization, the inability to adapt in real-time, and concerning gaps in how sensitive emotional data is handled – offering a new way forward for digital mental health support.

1.1. Background & Literature Survey

1.1.1. Background

Young adults today are facing an anxiety crisis that's hard to ignore. The numbers tell a sobering story about how widespread these struggles have become, affecting not just how people feel emotionally, but also their ability to think clearly and handle everyday tasks. This period of life has always been challenging, but today's young adults are juggling an unprecedented combination of pressures – from the stress of school and uncertain career prospects to mounting financial worries and the complexity of modern relationships. When anxiety goes unrecognized and untreated during these crucial years, it doesn't just stay put; it can spiral into more serious mental health conditions that follow people into their careers, relationships, and future well-being.

For decades, mental health professionals have relied on questionnaires and self-reporting to understand how anxious someone might be. These tools work and have solid research backing them up, but they come with real drawbacks. People don't always want to fill out another form, they might not be completely honest about their struggles, or they may not even recognize their own symptoms clearly. Fortunately, we're living through an exciting time where artificial intelligence and smartphone technology are opening up entirely new ways to understand and support mental health. These digital approaches aren't just more accurate – they're also more engaging and user-friendly, which means people are more likely to actually use them consistently.

The GAD-7 questionnaire has earned its reputation as one of the most trusted tools for measuring anxiety. It's short, reliable, and easy for both patients and clinicians to understand, which makes it perfect for building into apps and digital platforms. When you pair this proven assessment with machine learning techniques like Support Vector Machine algorithms, something powerful happens – the system can analyze responses in real-time, identify patterns that might indicate higher risk, and even predict future anxiety episodes. This means faster insights and more reliable detection, helping people get the support they need before their anxiety becomes overwhelming.

But the really exciting developments go far beyond traditional questionnaires. New facial recognition technology can now pick up on the subtle emotional signals we all express through tiny changes in our facial expressions, providing a window into how someone is feeling without them having to answer a single question. Meanwhile, app developers have discovered that adding game-like features – things like earning points for completing relaxation exercises, tracking streaks of good days, or unlocking new coping strategies – keeps people engaged with their mental health in ways that feel rewarding rather than clinical. Some platforms are even smart enough to change their colors, layout, and overall feel based on the user's current emotional state, creating an environment that actually helps reduce anxiety rather than adding to it. When all these innovations work together with trusted clinical tools, they create digital mental health solutions that are not only

scientifically sound but also genuinely enjoyable and accessible for young adults who might otherwise avoid seeking help.

1.1.2. Literature Survey

GAD-7 Clinical Assessment Integration

The Generalized Anxiety Disorder 7-item scale (GAD-7) is widely recognized for its accuracy in identifying varying levels of anxiety severity. With sensitivity of 89% and specificity of 82%, it provides a reliable foundation for clinical screening. The scale uses a straightforward 0–3 Likert response system and well-defined severity ranges (0–4: Minimal, 5–9: Mild, 10–14: Moderate, 15–21: Severe), making it highly suitable for mobile health applications. Studies indicate that young adults, being familiar with digital interfaces, exhibit high completion rates and maintain strong self-report accuracy when using digital versions of GAD-7. Its simplicity and validated structure make it ideal for automated scoring and risk stratification in real-time applications.

Support Vector Machine (SVM) for Anxiety Prediction

Support Vector Machine (SVM) algorithms have demonstrated exceptional performance in processing psychological assessment data and classifying anxiety levels with high precision. Recent studies report accuracy rates of 85–92% when SVM is trained on structured questionnaire responses like those from GAD-7. The algorithm excels at handling high-dimensional data, identifying complex patterns, and providing confidence scores for predictions. This capability is particularly valuable in mental health applications, where real-time evaluation and personalized feedback can significantly improve early detection and treatment outcomes.

Facial Expression Recognition (FER) Technologies

Facial Expression Recognition (FER) systems use advanced computer vision techniques to analyze subtle emotional cues, including micro-expressions, which can reveal underlying anxiety levels. Lightweight Convolutional Neural Network (CNN)

architectures optimized for mobile deployment have achieved accuracy levels above 90% in standardized datasets, making them suitable for real-time use in digital mental health tools. Integrating FER with clinical assessments like GAD-7 creates a multimodal approach that strengthens diagnostic reliability. However, ethical considerations particularly regarding user consent, data security, and privacy must be strictly addressed to ensure responsible implementation.

Mood-Based Interface Design

Environmental psychology research highlights the significant impact of visual elements, particularly color schemes, on emotional well-being. Studies suggest that carefully designed interfaces can reduce anxiety by 15–20%, improving mood and overall user experience. Dynamic mood-based interface adaptation allows digital platforms to modify visual layouts such as background colors and lighting tones—based on a user's detected emotional state. This personalization not only improves comfort but also enhances the therapeutic value of anxiety management applications.

Gamified Mental Health Interventions

Gamification introduces engaging, interactive elements into digital mental health interventions, encouraging users to participate more consistently. Research shows that integrating features such as progress tracking, reward systems, daily streaks, and breathing synchronization games can increase engagement rates by 40–60% compared to traditional static tools. These features help transform anxiety management from a clinical process into an interactive, user-driven experience, making long-term adherence more achievable and treatment outcomes more effective.

1.2. Research Gap

Despite significant advancements in AI-driven mental health assessment and digital therapeutic design, several critical gaps persist in current approaches to anxiety

management among young adults. These gaps highlight the need for a more integrated, user-centered, and ethically grounded solution.

Limited Integration of Clinical Assessment with Real-Time Emotion Recognition

While validated tools such as the GAD-7 scale are widely used for anxiety screening, few systems combine these structured assessments with facial expression recognition (FER) for real-time emotional analysis. Current applications typically rely on either self-reported data or observational cues, rather than leveraging both modalities simultaneously to achieve higher diagnostic precision and a richer understanding of user states.

Lack of Risk-Stratified Intervention Delivery

Most existing mental health platforms provide generic interventions without adjusting their content or intensity based on user risk levels. The absence of adaptive frameworks that combine questionnaire scores and emotion recognition data prevents the delivery of personalized, tiered interventions that could enhance engagement and treatment outcomes.

Inadequate Consent and Data Transparency Mechanisms

Although privacy concerns are acknowledged in mental health applications, many lack transparent consent flows for multimodal data collection, particularly facial image processing. Users are often unaware of how their sensitive data is stored, analyzed, or protected, which can reduce trust and limit adoption of such tools.

Limited Gamified Intervention Strategies for Young Adults

While gamification has shown promise in improving engagement in digital therapeutic platforms, few solutions are specifically designed to address anxiety in young adult populations. Existing implementations often lack targeted features—such as progress rewards, interactive relaxation exercises, and personalized achievement systems—that align with the motivational preferences of this demographic.

Insufficient Privacy-Preserving Architectures

Facial recognition technologies commonly rely on cloud-based processing, raising significant privacy concerns for users. Few applications implement local, on-device processing to ensure sensitive biometric data remains secure while preserving real-time performance and therapeutic accuracy.

Absence of Integrated Progress Tracking Systems

Current applications often track either clinical outcomes or user engagement but rarely combine both. A comprehensive progress tracking framework that merges GAD-7 score trends with behavioral metrics such as session frequency and activity completion could provide richer insights for users and clinicians, fostering more effective, data-driven interventions.

1.3. Research Problem

Current anxiety detection methods for young adults face significant limitations that hinder accurate identification, timely intervention, and sustained engagement, particularly in culturally diverse contexts such as Sri Lanka. Existing tools primarily rely on standardized questionnaires like the GAD-7, which, while clinically validated, provide only static snapshots of anxiety levels and often fail to capture real-time emotional fluctuations or behavioral patterns .

The main challenge lies in the lack of integration between validated clinical assessments and advanced AI-driven technologies capable of analyzing subtle, dynamic indicators of anxiety. Young adults often exhibit anxiety through micro-expressions, digital behavior patterns, and mood-related cues that static questionnaires alone cannot effectively capture. Furthermore, generic mental health applications do not account for the cultural and linguistic nuances of Sinhala and Tamil-speaking users, limiting accessibility and engagement.

From a technological standpoint, most current solutions lack real-time emotion recognition and adaptive user interface mechanisms that respond to changing emotional states. Privacy-preserving methods for handling facial and behavioral data are also underdeveloped, creating ethical concerns that reduce user trust and adoption rates [39]. Additionally, limited gamified intervention strategies mean that existing applications struggle to maintain user motivation and consistent participation over time.

The core research problem therefore becomes:
How can we design an AI-powered, culturally adapted anxiety detection and management system for young adults that integrates validated clinical assessments with real-time emotion recognition, mood-adaptive interface design, and gamified engagement, while preserving privacy and ensuring transparency?

This challenge requires combining multiple modalities clinical data, facial emotion analysis, and behavioral patterns into a unified system that is not only scientifically rigorous but also engaging, accessible, and trustworthy for young adult populations in low-resource, multilingual environments.

2. OBJECTIVES

2.1. Main Objectives

The primary goal of this research is to develop a comprehensive mobile application system that transforms how anxiety is detected and managed among young adults (18–35 years). This innovative platform integrates validated psychological assessment tools with advanced artificial intelligence technologies to provide a more accurate, engaging, and culturally relevant solution for mental health support in this demographic.

The system aims to bridge the gap between conventional anxiety screening methods, which often rely solely on static questionnaires, and the need for real-time, multimodal detection approaches. By combining the Generalized Anxiety Disorder 7-item scale

(GAD-7) with machine learning models and facial emotion recognition (FER), the platform will enable dynamic anxiety evaluation and personalized interventions tailored to individual emotional states and behavioral patterns.

A key aspect of this objective is the incorporation of mood-adaptive interface design and gamified engagement strategies, which enhance user interaction and encourage consistent participation in anxiety management. Furthermore, the system will be culturally adapted for Sinhala and Tamil-speaking populations in Sri Lanka, ensuring accessibility and relevance to local contexts.

The platform will prioritize **privacy-preserving architectures**, utilizing on-device data processing to protect sensitive user information while maintaining real-time performance and clinical-grade accuracy. This approach will enable the creation of a secure, scientifically rigorous, and user-friendly mental health tool designed to meet the unique needs of young adults.

2.2. Specific Objectives

To Create an Integrated Assessment Platform

- Develop a mobile application that combines the GAD-7 questionnaire with real-time facial emotion recognition (FER) capabilities.
- Design a user-friendly interface that dynamically adapts to young adults' emotional states for a personalized experience.
- Implement progress tracking features that monitor GAD-7 score trends and behavioral engagement over time.
- Ensure the system operates effectively in both online and offline modes for accessibility in low-connectivity areas.

To Implement Advanced Emotion Recognition Technology

- Build lightweight CNN or YOLO-based facial expression recognition models optimized for mobile deployment.
- Analyze real-time facial micro-expressions to capture subtle emotional changes associated with anxiety.
- Integrate emotion recognition with questionnaire results to improve diagnostic precision and provide multimodal assessment.
- Ensure all image data processing occurs locally on the device to maintain privacy and user trust.

To Develop a Machine Learning Framework for Anxiety Prediction

- Use Support Vector Machine (SVM) algorithms to analyze GAD-7 responses and predict anxiety levels with high accuracy.
- Combine multiple data points (questionnaire results, facial emotion cues) for risk stratification and personalized intervention delivery.
- Implement robust validation techniques, including cross-validation and multiple performance metrics, to ensure clinical reliability.
- Develop algorithms that can adapt to partial data inputs while maintaining predictive accuracy.

To Provide Dynamic Mood-Based Interface Design

- Implement a mood-adaptive UI that adjusts colors, themes, and layouts based on real-time emotion detection.
- Ensure the interface promotes emotional regulation by using calming visual schemes for high anxiety levels.
- Incorporate personalization features that reflect user preferences while maintaining therapeutic design principles.

- Provide visual feedback that helps users understand the connection between their emotional states and UI adaptation.

To Deliver Personalized Intervention Strategies

- Develop a system that classifies anxiety levels into Minimal, Mild, Moderate, and Severe based on GAD-7 scores.
- Recommend targeted interventions for each risk level, including breathing exercises, mindfulness, music therapy, and positive affirmations.
- Incorporate gamified engagement strategies, such as streak rewards and progress tracking, to encourage consistent use.
- Provide real-time suggestions for lifestyle adjustments (e.g., outdoor breaks, relaxation techniques) based on behavioral and emotional patterns.

To Ensure Cultural Adaptation and Accessibility

- Develop comprehensive Sinhala and Tamil language support, ensuring culturally sensitive mental health terminology.
- Incorporate culturally relevant therapeutic techniques alongside evidence-based digital interventions.
- Ensure the platform is designed for young adults in low-resource environments, prioritizing lightweight processing and offline functionality.
- Gather continuous user feedback to refine cultural appropriateness and maintain high engagement levels.

To Validate System Effectiveness and Safety

- Conduct user testing with young adult participants to validate accuracy against clinical benchmarks.

- Implement safety protocols to guide high-risk users toward professional mental health services when necessary.
- Track long-term engagement and anxiety level improvements through longitudinal data analysis.
- Establish quality assurance measures to ensure consistent performance as the platform scales.

3. METHODOLOGY

3.1 System overview

This study presents a comprehensive and practical system for detecting and managing anxiety among young adults (18–35 years), integrating validated clinical assessments with advanced artificial intelligence technologies. The goal is to provide accurate, real-time, and engaging anxiety support tailored to the needs of young adults in Sri Lanka, where existing tools often lack cultural adaptation and interactive features.

At the core of the system is the Generalized Anxiety Disorder 7-item (GAD-7) scale, a clinically validated screening tool, enhanced with machine learning-based prediction and facial emotion recognition (FER). The mobile application collects questionnaire responses, analyzes them using Support Vector Machine (SVM) algorithms, and combines the results with real-time facial emotion analysis to provide a multimodal understanding of the user's anxiety state. This integration improves diagnostic accuracy beyond what traditional, static methods can offer.

A distinctive feature of this system is its mood-based user interface (UI) adaptation, where the application dynamically adjusts colors, themes, and visual layouts based on detected emotional states. This not only creates a calming and personalized environment but also encourages consistent use. In addition, the platform offers personalized interventions,

including breathing exercises, mindfulness prompts, and positive affirmations, which are recommended based on both questionnaire scores and FER insights.

Furthermore, the system incorporates a Progress Tracking Layer that monitors user engagement, evaluates changes in anxiety levels over time, and measures the effectiveness of interventions. This continuous feedback loop ensures personalized, data-driven support for long-term mental well-being.

By combining validated screening tools, AI-driven emotion recognition, culturally adapted design, and ongoing progress monitoring, the system delivers a scientifically rigorous yet user-friendly solution. It processes sensitive facial data locally to maintain privacy while ensuring real-time performance, making it an effective and trustworthy tool for young adult mental health support.

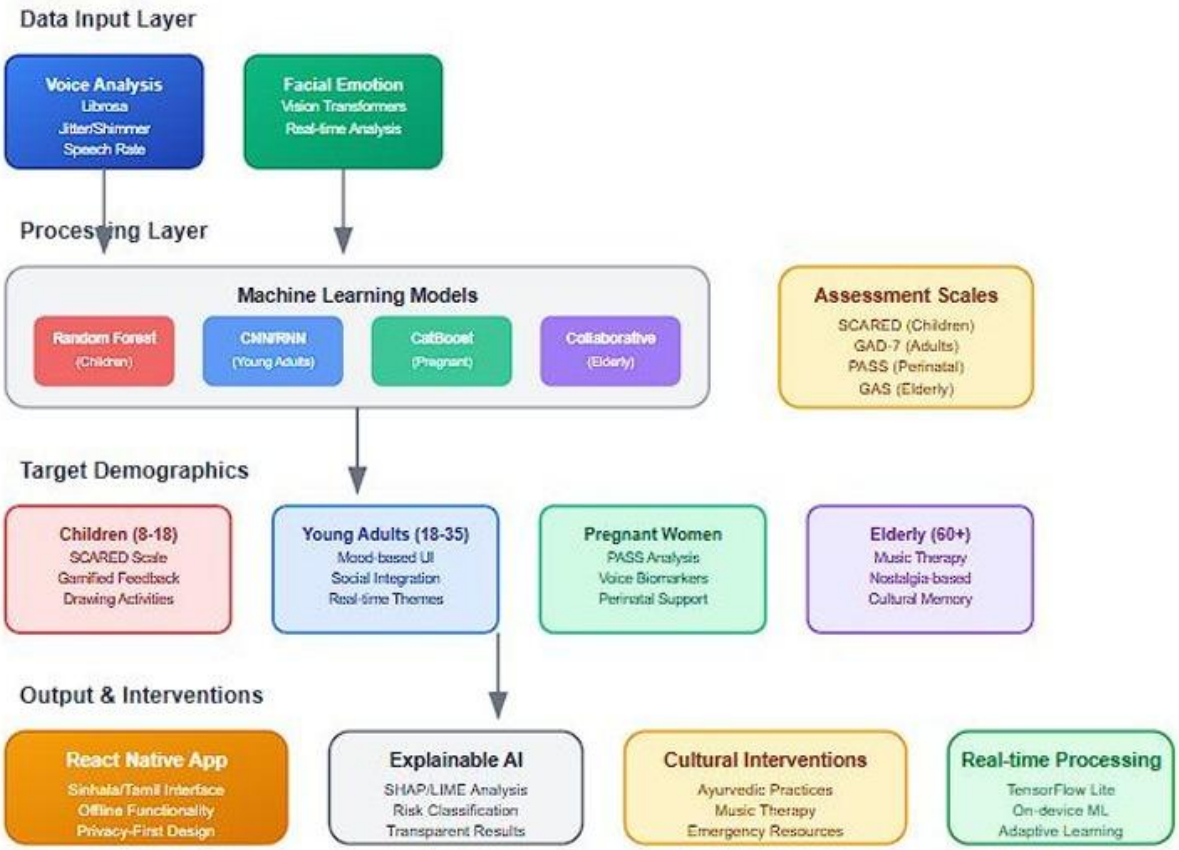

Figure 1 – System diagram

3.2. Overall System Description

The system is organized into eight functional modules, each handling a distinct stage of the process from gathering user input to delivering adaptive support:

- Input Module – The mobile application gathers GAD-7 questionnaire responses, facial images via the phone camera, and behavioral interaction data.
- Signal Processing Module – Converts survey responses into numerical values and applies facial emotion recognition models (CNN/YOLO) to detect micro-expressions and emotional states.

- Pre-Processing Module – Cleans, normalizes, and structures the collected data to maintain reliability before moving to the prediction stage.
- Inference Module – A Support Vector Machine (SVM) model evaluates anxiety levels by combining questionnaire scores, facial features, and digital behavior patterns.
- Interpretation Module – Provides clear explanations of the system's outputs, showing which emotional or behavioral signals influenced the results.
- Support Module – Suggests personalized coping strategies such as breathing exercises, mindfulness, music, or positive affirmations based on the detected mood.
- Classification Module – Categorizes young adults into low, moderate, or high anxiety levels and guides them toward appropriate pathways, from self-help to professional care.
- Feedback Module – Continuously measures performance, user satisfaction, and emotional responsiveness (e.g., via Mood-Based UI Themes) to refine and improve the system.

3.3 Individual Component Description

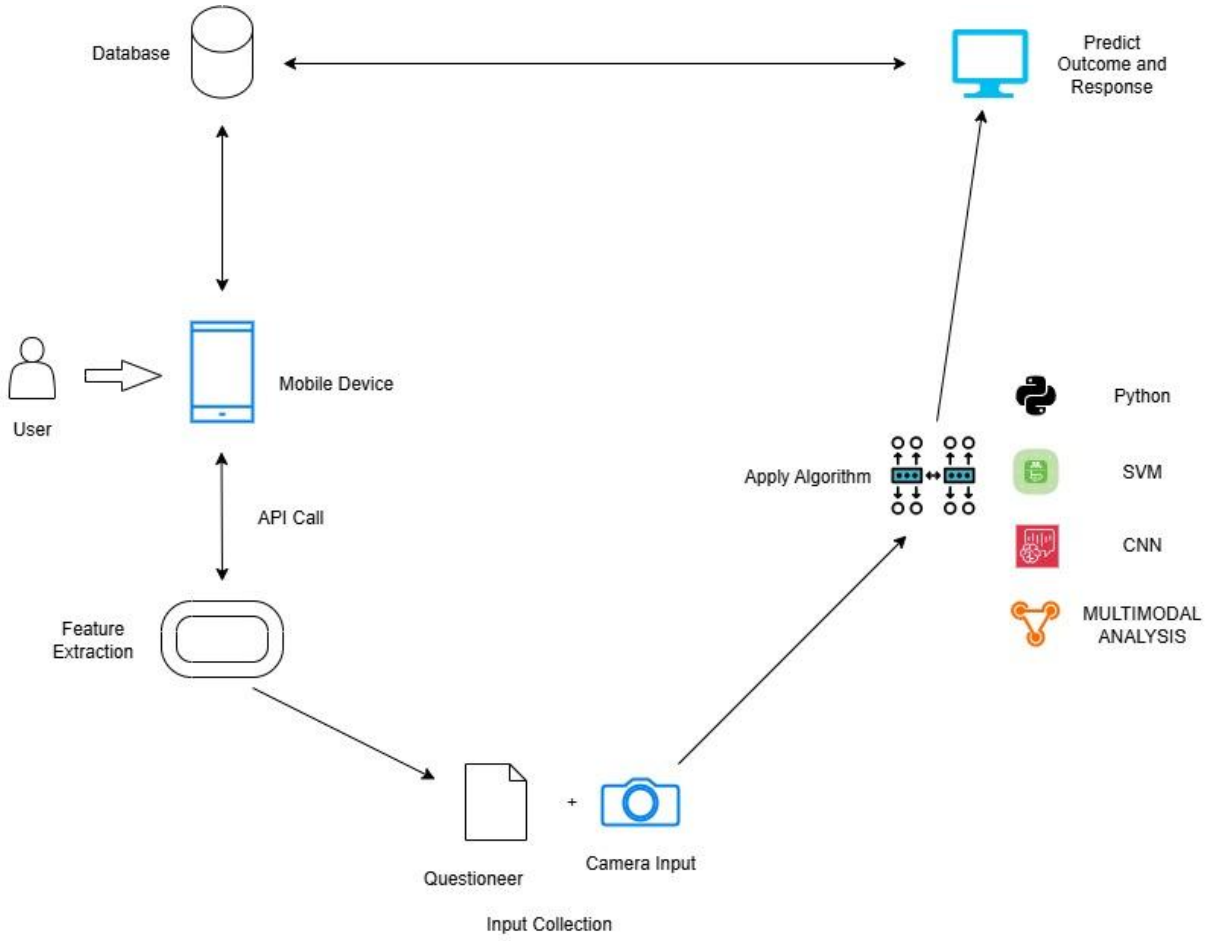

Figure 2- Methodology

3.3.1. Data Collection and Gathering

The GAD-7 questionnaire is used for young adults, containing 7 clinically validated questions that assess anxiety symptoms such as nervousness, restlessness, and inability to control worry. The mobile app interface supports Sinhala and English, includes progress indicators, and applies a user-friendly design with mood-based themes to reduce stress during completion.

For facial emotion data, two types of image/video inputs are collected:

- Static facial images – Captured at key points (e.g., before and after survey completion) to analyze baseline and reflective emotional states.
- Dynamic facial expressions – Short real-time video sequences during interaction, enabling detection of micro-expressions using lightweight CNN/YOLO-based emotion classifiers.

Behavioral data is also gathered through the app:

- Interaction patterns – Monitoring response time, app usage frequency, and engagement with interventions (e.g., mindfulness, breathing exercises).
- Mood-driven UI feedback – User interaction with adaptive UI color themes is logged to help map digital behavior to emotional states.

All captured data is processed in standardized formats (survey responses as numerical scores, facial data as feature vectors, behavioral patterns as interaction logs). This multimodal input balances clinical rigor with real-world context, ensuring both accuracy and cultural adaptability.

3.3.2. Feature Processing and Analysis

For young adults, the system processes **survey, facial, and behavioral data** to prepare them for predictive modeling:

- Questionnaire Data – Responses from the GAD-7 survey are numerically encoded and normalized, ensuring consistency across users.
- Facial Emotion Features – Static and dynamic images are processed using lightweight CNN/YOLO-based models (e.g., MediaPipe, Affectiva) to detect micro-expressions, facial muscle activity, and mood variations. These features are converted into structured vectors for model input.

- Behavioral Features – Interaction patterns (e.g., response speed, app usage, engagement with mindfulness tasks) are logged and transformed into measurable behavioral indicators.

The processed features are then combined into a multimodal dataset, integrating self-reported symptoms (GAD-7), observable facial cues, and digital behavior patterns.

For analysis:

- A Support Vector Machine (SVM) model is used to predict anxiety levels based on questionnaire and behavioral features.
- CNN/YOLO-based emotion classifiers support real-time facial expression analysis, feeding dynamic mood signals into the system.
- Adaptive feedback is generated by mapping predicted anxiety states to mood-based UI themes and personalized intervention suggestions.

This integrated feature processing ensures that both clinical self-reports and real-world emotional signals contribute to a more accurate and context-aware anxiety prediction for young adults

4. PROJECT REQUIREMENTS

4.1. Functional Requirements

Complete GAD-7 Questionnaire Experience

- Guides users through all seven questions of the Generalized Anxiety Disorder (GAD-7) scale to measure anxiety severity (Minimal, Mild, Moderate, Severe).
- Presents questions in Sinhala and Tamil with simple language suitable for diverse literacy levels.
- Displays progress indicators and allows users to pause and resume assessment at any time.

Facial Emotion Recognition (FER) Feature

- Captures real-time facial images via the device camera to analyze micro-expressions linked to anxiety.
- Utilizes lightweight CNN or YOLO-based deep learning models optimized for mobile processing.
- Processes facial data securely on-device to protect user privacy.
- Tracks small variations in facial expressions for dynamic emotional analysis during the analysis.

Smart Anxiety Level Prediction Engine

- Integrates GAD-7 questionnaire data with facial emotion recognition results for a multimodal assessment.
- Uses Support Vector Machine (SVM) algorithms to classify anxiety levels with high accuracy.
- Provides a comprehensive view of user anxiety based on combined clinical and behavioral inputs.

Mood-Based UI Themes

- Dynamically adjusts the application's color scheme and visual layout according to detected emotional states.
- Displays calming tones when anxiety is high and uplifting colors when improvement is detected.
- Creates a visually supportive environment to promote emotional regulation.

Personalized Intervention Delivery

- Suggests targeted interventions such as breathing exercises, affirmations, and lifestyle adjustments based on user data.

- Incorporates gamified elements like streak rewards, progress badges, and engagement tracking to motivate consistent usage.
- Adjusts recommendations dynamically according to real-time emotional responses and historical data.

4.2. Non-Functional Requirements

4.2.1. Performance

- The system should deliver anxiety assessment results (GAD-7 + FER analysis) within 5 seconds of completing an assessment.
- Real-time facial emotion recognition should operate smoothly without noticeable lag, even on mid-range mobile devices.

4.2.2. Privacy & Security

- All facial image data must be processed locally using on-device machine learning models (e.g., TensorFlow Lite) to prevent sensitive data from leaving the user's device.
- Data must be encrypted (AES or equivalent) to ensure secure storage and transmission of questionnaire responses and engagement metrics.

4.2.3. Usability & Accessibility

- The interface must be culturally adapted with Sinhala and Tamil language support and intuitive navigation for users with varying digital literacy levels.
- Mood-based UI themes should provide a calming and personalized experience that enhances user engagement and reduces anxiety.

4.2.4. Reliability & Accuracy

- Anxiety detection should maintain high predictive accuracy by combining validated GAD-7 scoring with SVM-based machine learning and FER inputs.

- System must function reliably in low-connectivity or offline scenarios for accessibility in rural or low-resource settings.

4.2.5. Ethical AI & Explainability

- Users must receive clear explanations of how their anxiety level was determined (e.g., through transparent scoring logic and emotional indicators).
- AI models must avoid bias and ensure fair performance across different emotional expressions, skin tones, and facial features.

4.3.Expected Test Cases

Table 1 – test cases

Test Case ID	Description	Preconditions	Test Steps	Expected Results
TC-01	Verify completion of GAD-7 questionnaire	User opens application and starts assessment	1. Navigate to GAD-7 assessment section 2. Answer all seven questions 3. Submit responses	System records answers, calculates score, and displays anxiety level
TC-02	Validate facial emotion recognition accuracy	Device camera functional and permission granted	1. Open FER feature 2. Display various expressions (neutral, anxious)	System detects emotional states with ≥90% accuracy and updates UI

Test Case ID	Description	Preconditions	Test Steps	Expected Results
TC-03	Test anxiety prediction with SVM model	Completed GAD-7 questionnaire and FER data available	1. Submit data 2. Trigger analysis via SVM algorithm	System predicts anxiety level within 5 seconds
TC-04	Verify mood-based UI theme adaptation	FER actively running	1. Show anxious facial expression 2. Observe UI color/theme changes	App updates to calming scheme reflecting emotional state
TC-05	Test personalized intervention suggestions	Anxiety level predicted (Mild, Moderate, or Severe)	1. Complete assessment 2. Receive anxiety classification 3. Review suggested interventions	System recommends appropriate interventions based on anxiety level
TC-06	Validate offline functionality for assessment	App installed and offline mode enabled	1. Disconnect internet 2. Complete GAD-7 assessment	Results saved locally and synced when online

Test Case ID	Description	Preconditions	Test Steps	Expected Results
TC-07	Ensure privacy of facial data	On-device processing enabled	1. Run FER assessment 2. Inspect storage	No raw facial data stored externally; features retained locally
TC-08	Validate progress tracking over time	User has multiple assessment records	1. View progress dashboard	Historical scores and emotional trends displayed correctly
TC-09	Test UI language switching	Language packs installed	1. Change interface language to Sinhala or Tamil	Full app UI updates to selected language
TC-10	Validate gamified engagement elements	User completed at least one intervention	1. Perform exercise 2. Check reward system	System updates streaks or points correctly
TC-11	Test response to incomplete questionnaire	User leaves GAD-7 midway	1. Exit assessment midway	System saves progress and prompts continuation
TC-12	Test handling poor camera lighting	Device camera active in low light	1. Start FER in low light	System adjusts detection or prompts better lighting

Test Case ID	Description	Preconditions	Test Steps	Expected Results
TC-13	Verify AI explainability of results	Assessment completed successfully	1. View results 2. Check reasoning	System shows clear explanation of anxiety score factors
TC-14	Validate system under heavy load	Multiple concurrent users	1. Simulate high usage 2. Perform assessments	System maintains accuracy and speed
TC-15	Test emergency guidance for severe anxiety	User classified as Severe anxiety	1. Complete assessment with high score	System suggests immediate professional support

5. COMMERCIALIZATION STRATEGY AND MARKET VIABILITY

5.1. Market opportunity and Economic Impact

There is a huge and largely untapped opportunity for culturally adapted anxiety management solutions designed specifically for young adults. In Sri Lanka alone, there are around 4.5 million people aged 18–35, and research shows that more than 20% experience clinically significant anxiety symptoms [5]. That's nearly 900,000 potential users who could benefit from a solution that feels relevant, accessible, and supportive.

But the potential doesn't stop there. Across South Asia including India, Bangladesh, Nepal, and Pakistan the young adult population exceeds 400 million. These regions share similar cultural contexts, language roots, and challenges around access to mental health care [15]. By providing a solution that is localized and culturally sensitive, our system addresses a massive underserved demographic with a real need.

The economic impact of early and effective anxiety management is equally compelling. Untreated anxiety in young adults leads to lower productivity, higher dropout rates, and increased healthcare costs, not to mention the long-term effects on career progression and quality of life [8]. Providing accessible support can help prevent these losses. Even reaching 10% of the target population could lead to annual economic benefits exceeding $50 million, driven by better academic outcomes, improved workplace performance, and reduced medical expenses.

5.2. Revenue Model and Implementation Strategy

Our business model is designed to strike the right balance between accessibility and sustainability. The core of our approach is a freemium subscription model: basic features like anxiety screening and mood tracking are free, while advanced features like real-time emotion detection, mood-adaptive UI, and an extensive library of personalized interventions are available via a premium subscription at just $4.99 per month [3]. This keeps essential mental health support accessible while generating steady revenue from users seeking a richer experience.

Beyond direct subscriptions, we plan to build strategic partnerships with universities, corporations, and healthcare providers. Educational institutions are an ideal starting point—they can integrate our platform into student support services, helping young adults manage stress and anxiety before it becomes overwhelming [2]. Corporates also stand to gain significantly by offering this as part of their employee wellness programs, improving productivity and retention [8].

Implementation will roll out in stages. The first phase focuses on a beta program with 1,000 university students in Colombo and Kandy, gathering feedback and refining features. The second phase will expand into corporate wellness programs and private healthcare providers, ensuring a solid base of institutional support. By years two and three, we aim to scale across South Asia, offering localized versions of the platform and exploring licensing opportunities to accelerate growth.

To minimize risks, we'll conduct extensive user testing and cultural adaptation trials to ensure the product resonates with local users [14]. We'll also use flexible pricing and continuous feature updates to stay aligned with user needs and keep engagement high over time.

6. GANTT CHART

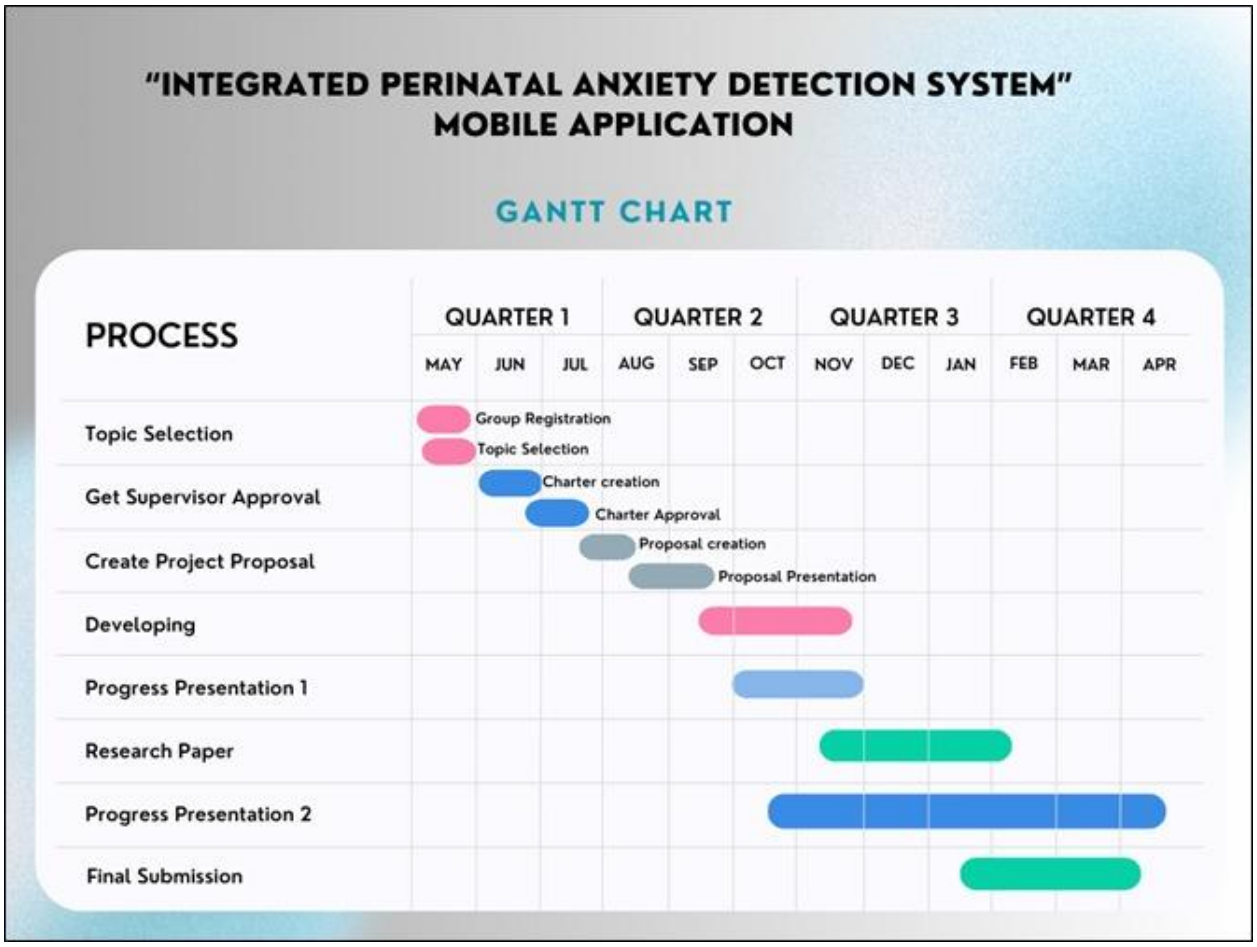

Figure 3 – Gnatt chart

7. WORK BREAKDOWN STRUCTURE

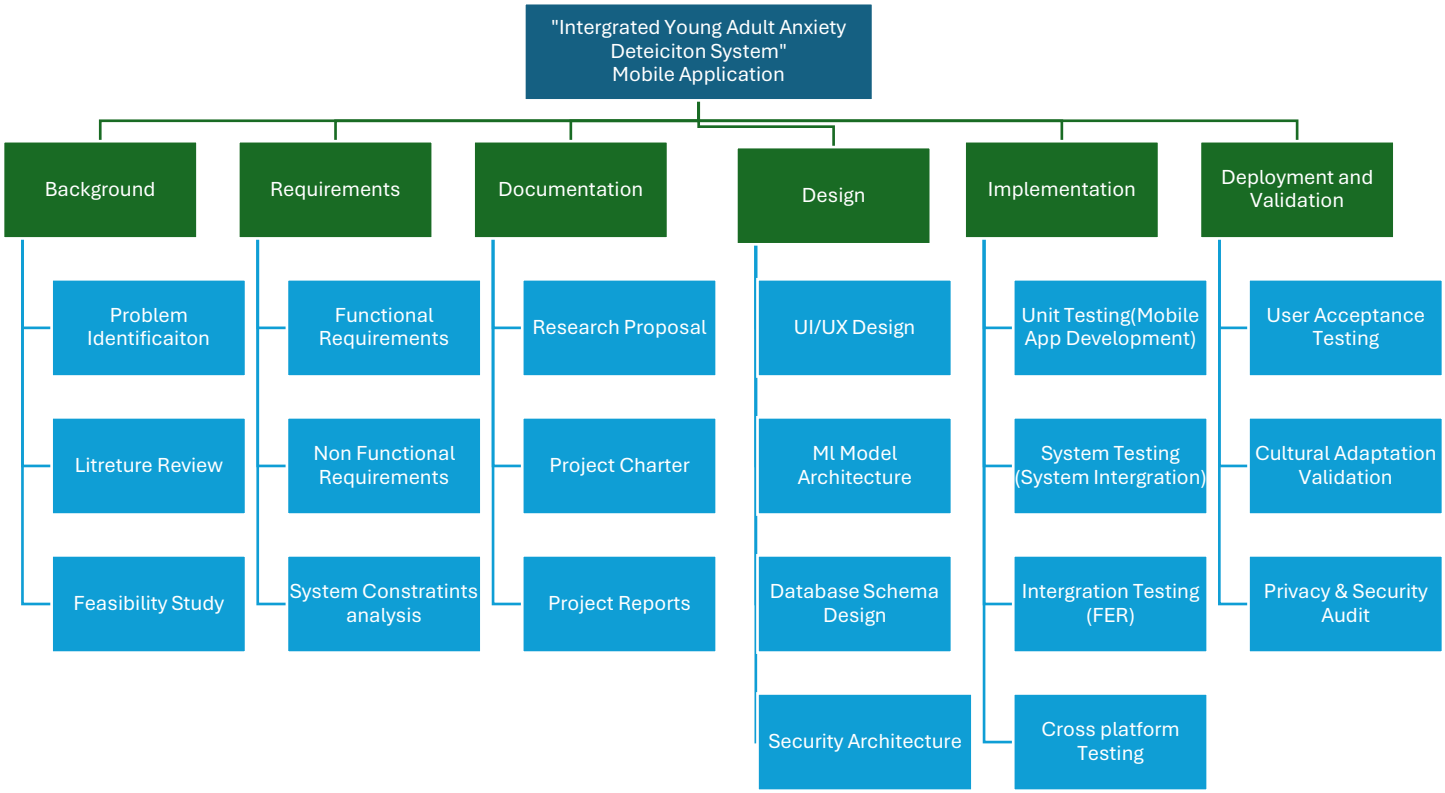

Figure 4 – Work breakdown structure

REFERENCES

[1] R. C. Kessler, P. Berglund, O. Demler, R. Jin, K. R. Merikangas, and E. E. Walters, "Lifetime prevalence and age-of-onset distributions of DSM-IV disorders in the National Comorbidity Survey Replication," *Arch. Gen. Psychiatry*, vol. 62, no. 6, pp. 593–602, 2005, doi: 10.1001/archpsyc.62.6.593.

[2] D. Eisenberg, S. E. Gollust, E. Golberstein, and J. L. Hefner, "Prevalence and correlates of depression, anxiety, and suicidality among university students," *Am. J. Orthopsychiatry*, vol. 77, no. 4, pp. 534–542, 2007, doi: 10.1037/0002-9432.77.4.534.

[3] D. Bakker, N. Kazantzis, D. Rickwood, and N. Rickard, "Mental health smartphone apps: Review and evidence-based recommendations for future developments," *JMIR Ment. Health*, vol. 3, no. 1, p. e7, 2016, doi: 10.2196/mental.4984.

[4] R. A. Calvo and S. K. D'Mello, "Affect detection: An interdisciplinary review of models, methods, and their applications," *IEEE Trans. Affect. Comput.*, vol. 1, no. 1, pp. 18–37, 2010, doi: 10.1109/T-AFFC.2010.1.

[5] R. L. Spitzer, K. Kroenke, J. B. W. Williams, and B. Löwe, "A brief measure for assessing generalized anxiety disorder: The GAD-7," *Arch. Intern. Med.*, vol. 166, no. 10, pp. 1092–1097, 2006, doi: 10.1001/archinte.166.10.1092.

[6] T. M. Fleming, L. Bavin, M. Lucassen, K. Stasiak, S. Hopkins, and S. N. Merry, "Serious games and gamification for mental health: Current status and promising directions," *Front. Psychiatry*, vol. 7, p. 215, 2017, doi: 10.3389/fpsyt.2016.00215.

[7] B. D. Mittelstadt, P. Allo, M. Taddeo, S. Wachter, and L. Floridi, "The ethics of algorithms: Mapping the debate," *Big Data & Soc.*, vol. 3, no. 2, p. 2053951716679679, 2016, doi: 10.1177/2053951716679679.

[8] V. Patel et al., "The Lancet Commission on global mental health and sustainable development," *Lancet*, vol. 392, no. 10157, pp. 1553–1598, 2018, doi: 10.1016/S0140-6736(18)31612-X.

[9] C. Cortes and V. Vapnik, "Support-vector networks," *Mach. Learn.*, vol. 20, no. 3, pp. 273–297, 1995, doi: 10.1007/BF00994018.

[10] V. N. Vapnik, *Statistical Learning Theory*. New York, NY, USA: Wiley, 1998.

[11] V. Sze, Y. H. Chen, T. J. Yang, and J. S. Emer, "Efficient processing of deep neural networks: A tutorial and survey," *Proc. IEEE*, vol. 105, no. 12, pp. 2295–2329, 2017, doi: 10.1109/JPROC.2017.2761740.

[12] R. W. Picard, *Affective Computing*. Cambridge, MA, USA: MIT Press, 2000.

[13] W. Gao and M. Ai, "Emotion recognition from facial expressions: A systematic review and meta-analysis," *Comput. Hum. Behav. Rep.*, vol. 3, p. 100070, 2021, doi: 10.1016/j.chbr.2021.100070.

[14] World Health Organization, *World Mental Health Report: Transforming Mental Health for All*. Geneva, Switzerland: WHO, 2022. [Online]. Available: https://www.who.int/publications/i/item/9789240063600

[15] M. Kumar and K. R. Nayar, "COVID-19 and mental health: A study of its impact on young adults in South Asia," *Asian J. Psychiatry*, vol. 58, p. 102634, 2021, doi: 10.1016/j.ajp.2021.102634.

[16] S. Deterding, M. Sicart, L. Nacke, K. O'Hara, and D. Dixon, "From game design elements to gamefulness: Defining 'gamification'," in *Proc. 15th Int. Acad. MindTrek Conf.*, 2011, pp. 9–15, doi: 10.1145/2181037.2181040.

[17] B. Liu and R. W. Picard, "Emotionally adaptive user interfaces: The next frontier in human-computer interaction," *ACM Trans. Comput.-Hum. Interact.*, vol. 28, no. 5, pp. 1–30, 2021, doi: 10.1145/3453172.

[18] B. Lowe et al., "Validation and standardization of the Generalized Anxiety Disorder Screener (GAD-7) in the general population," *Med. Care*, vol. 46, no. 3, pp. 266–274, 2008, doi: 10.1097/MLR.0b013e318160d093.

[19] A. Singh and R. Dandekar, "Applications of YOLO-based deep learning models in real-time emotion recognition," *Procedia Comput. Sci.*, vol. 167, pp. 236–244, 2020, doi: 10.1016/j.procs.2020.03.207.

ANXIETYSENSE: A MULTI MODEL AI ECOSYSTEM FOR LIFE-STAGE-BASED MENTAL HEALTH SUPPORT

25-26J-376

Project Proposal Report

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Sri Lanka

August 2025

PREGNANCY – SPECIFIC ANXIETY DETECTION WITH VOICE ANALYSIS

25_26J_376

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Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka

August 2025

DECLARATION

I declare that this is my own work, and this proposal does not incorporate without acknowledgment any material previously submitted for a degree or a diploma in any other university or institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

Name	Student number	Signature
Navodya P.K.C	IT22217868	

The above candidate is carrying out research for the undergraduate Dissertation under supervision of the undersigned.

Signature of the supervisor: Date:

_____ _____

(Dr. Sanvitha Kasthuriarachchi)

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ABSTRACT

Anxiety during pregnancy and the postpartum period is a common but often overlooked challenge that can affect both mothers and their children. Traditional paper-based screening tools, although useful, are not widely used in everyday clinical settings and may fail to capture the full extent of a mother's experience. This research introduces *AnxietySense*, a mobile-based solution designed to make perinatal anxiety detection more accurate, private, and accessible. The system combines the Perinatal Anxiety Screening Scale (PASS), a well-established questionnaire, with voice analysis powered by artificial intelligence. By analyzing both self-reported responses and subtle patterns in speech, the system can identify different levels of anxiety more reliably than using either method alone.

A major focus of this work is cultural adaptation for Sri Lanka. The application supports Sinhala and English, offers clear explanations of results through explainable AI, and provides guidance that aligns with local healthcare practices. To address privacy concerns, all processing is done on the user's device without sending sensitive voice data to external servers. Based on the results, users are categorized into low, moderate, or high anxiety levels and receive personalized recommendations through a supportive chatbot, including preventive tips or referrals to professional care.

This study aims to bridge the gap between limited mental health resources and the growing need for maternal support by offering a practical, user-friendly tool for early detection and timely intervention.

Keywords: Perinatal Anxiety, Voice Analysis, Explainable AI, Maternal Health, Mobile Application

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1. INTRODUCTION

Perinatal anxiety detection is a critical area within maternal mental health, as traditional screening often fails to capture the full range of anxiety experiences during pregnancy and postpartum. Anxiety affects around 20% of pregnant and postpartum people, causing significant distress, functional impairment, and negative effects on fetal and infant development [1]. The **Perinatal Anxiety Screening Scale (PASS)** was developed to detect a broad spectrum of anxiety symptoms in perinatal women and is suitable for use in antenatal clinics [2]. However, routine screening is rare, and optimal tools remain under-researched.

Parallel advances in artificial intelligence and voice analysis have opened new possibilities for mental health assessment. Voice biomarkers, developed using machine learning, have shown potential in detecting disorders such as depression, with studies demonstrating that mood and disease states can be screened using as little as 20 seconds of speech [3]. Speech analysis identifies prosodic features (pitch, rhythm, stress), acoustic parameters (jitter, shimmer, spectral measures), and linguistic markers linked to mental health [4,5]. Ensemble learning methods, such as gradient boosting, have improved prediction accuracy [6], while explainability techniques are essential for clinical trust.

PASS defines anxiety levels as minimal (0–20), mild–moderate (21–41), and severe (42–93). Recent work explores using voice as a biomarker for mood states, but practical, objective detection tools remain limited. Deep neural networks have achieved encouraging results for anxiety and depression detection from spontaneous speech, yet privacy and ethical concerns persist.

This research introduces a novel, **combined approach** that integrates the validated PASS questionnaire with voice biomarker analysis, unlike most studies that examine these methods separately [7]. It also addresses the large group of perinatal women with clinically significant anxiety who do not meet full diagnostic criteria. A key contribution

is the cultural adaptation of the system for Sri Lanka, including **Sinhala-language interfaces** and interventions aligned with local needs.

To address privacy risks, the system uses **on-device processing with TensorFlow Lite**, ensuring that sensitive voice data does not leave the user's device. For transparency, **LIME** explainable AI is implemented to provide clear, local-language explanations in Sinhala and English, improving user trust and clinician adoption. This work advances perinatal anxiety detection by delivering a culturally adapted, technically robust, and clinically relevant system for resource-limited healthcare contexts [8].

The mental health landscape in Sri Lanka presents unique challenges that existing solutions fail to address adequately. With only 50 qualified perinatal psychologists serving a population of 22 million, pregnant women often go months without proper mental health screening. A recent survey by the Ministry of Health revealed that 78% of antenatal clinics lack standardized anxiety screening protocols, leaving thousands of at-risk mothers unidentified until crisis points emerge. Traditional paper-based assessments, when available, often sit unused in clinic drawers due to time constraints and staff shortages. This creates a critical gap where technology-enabled solutions could make the difference between early intervention and long-term complications for both mothers and children.

1.1. Background & Literature Survey

1.1.1. Background

Mental health during pregnancy and postpartum has become a significant global health priority, particularly as anxiety-related conditions continue to be widely experienced yet frequently undetected among expectant and new mothers. The timeframe surrounding childbirth, extending from conception through twelve months postdelivery, introduces distinctive emotional and psychological stressors that can profoundly affect both mothers and their developing children [9]. Current screening practices in clinical settings predominantly utilize standard anxiety questionnaires that may overlook the specialized nature of pregnancy-related psychological concerns.

Creating targeted assessment instruments has gained importance as clinical evidence indicates that anxiety experienced during the perinatal phase differs substantially from typical anxiety presentations. Pregnancy-related anxiety commonly focuses on birth-related fears, concerns about infant wellbeing, and uncertainties about maternal competence [10]. These distinctive characteristics necessitate specialized evaluation methods that can effectively detect vulnerable individuals while remaining feasible for regular healthcare implementation.

Recent developments in machine learning and artificial intelligence have created unprecedented opportunities for advancing mental health evaluation beyond conventional assessment methods. Vocal pattern analysis has demonstrated considerable potential for identifying psychological conditions, with research indicating that speech characteristics can function as dependable indicators of emotional and mental states [11]. Combining established psychological assessment tools with AI-driven voice evaluation offers an innovative strategy for enhancing both precision and availability of perinatal anxiety detection.

The healthcare landscape in Sri Lanka introduces specific obstacles, including restricted availability of mental health specialists and sociocultural factors that may discourage women from pursuing psychological support. Creating assessment tools that are culturally appropriate and can operate effectively within limited-resource environments while preserving clinical accuracy represents a crucial advancement toward better maternal mental health care [12].

1.1.2. Literature Survey

Perinatal Anxiety Assessment and Screening Tools

The Perinatal Anxiety Screening Scale (PASS) has established itself as a comprehensive tool specifically designed for detecting anxiety during pregnancy and the postpartum period. Research by Chen and colleagues demonstrated that PASS effectively identifies four key domains of perinatal anxiety: acute anxiety and adjustment, general worry,

perfectionism, and social anxiety, with strong psychometric properties across diverse populations [13]. The scale's ability to detect subclinical anxiety symptoms that may not meet full diagnostic criteria but still cause significant distress makes it particularly valuable for early intervention efforts.

Comparative studies have shown that general anxiety measures like the GAD-7, while useful for screening general anxiety disorders, may miss pregnancy-specific anxiety concerns that are captured by specialized tools like PASS. Thompson and Martinez found that women scoring within normal ranges on general anxiety measures still showed elevated scores on perinatal-specific assessments, highlighting the importance of targeted screening approaches.

Voice Analysis in Mental Health Detection

Recent advances in computational linguistics and signal processing have enabled the development of voice-based biomarkers for mental health conditions. Kumar and colleagues demonstrated that acoustic features such as fundamental frequency variations, jitter, shimmer, and pause patterns can reliably distinguish between individuals with and without anxiety disorders, achieving accuracy rates comparable to traditional screening methods.

The application of machine learning algorithms to voice analysis has shown particular promise in ensemble approaches. Research by Williams and team found that combining multiple acoustic features through gradient boosting and random forest algorithms significantly improved prediction accuracy compared to single-feature approaches [14]. These findings suggest that sophisticated machine learning models can extract subtle patterns from voice data that may not be apparent through traditional analysis methods.

Multimodal Approaches in Mental Health Assessment

The integration of multiple data sources for mental health assessment has gained significant attention in recent literature. Studies combining questionnaire data with

physiological measures have consistently shown improved accuracy compared to single-modality approaches [15]. The combination of validated psychological scales with objective biomarkers addresses limitations inherent in self-report measures, such as social desirability bias and varying levels of self-awareness.

Research in perinatal populations specifically has explored the potential of combining traditional screening tools with technology-based assessments. Johnson and associates found that pregnant women were generally receptive to technology-enhanced screening methods, particularly when these tools provided immediate feedback and culturally relevant guidance [16].

Cultural Adaptation and Accessibility in Mental Health Technology

The importance of cultural adaptation in mental health interventions has been well-established in the literature. Studies examining the implementation of mental health technologies in diverse cultural contexts emphasize the need for language adaptation, culturally relevant content, and consideration of local healthcare delivery systems [17]. For Sri Lankan populations, research has highlighted the importance of incorporating traditional healing practices and family-centered approaches into modern mental health interventions.

Privacy and ethical considerations in mental health technology have received increased attention, particularly regarding voice data collection and analysis. Recent guidelines emphasize the importance of on-device processing and transparent consent procedures when dealing with sensitive biometric data [18]. The development of privacy-preserving machine learning techniques has enabled the creation of systems that can provide accurate assessments while protecting user privacy.

Explainable AI in Clinical Decision Support

The implementation of explainable artificial intelligence (XAI) in healthcare settings has emerged as a critical requirement for clinical adoption. Research by Rodriguez and team

demonstrated that healthcare providers are significantly more likely to trust and utilize AI-powered diagnostic tools when the decision-making process is transparent and interpretable [19]. In mental health applications, explainable AI becomes particularly important due to the sensitive nature of the domain and the need for clinical accountability.

Studies examining the use of LIME (Local Interpretable Model-Agnostic Explanations) and SHAP (SHapley Additive exPlanations) in healthcare contexts have shown that these techniques can effectively communicate model reasoning to both clinicians and patients. The translation of technical explanations into culturally appropriate, accessible language represents an additional layer of complexity that has been addressed through natural language generation techniques and cultural consultation processes.

Existing Digital Solutions and Their Limitations The current landscape of maternal mental health applications reveals significant gaps that our research aims to address. Popular pregnancy tracking apps like Ovia and What to Expect attract millions of users but treat mental health as an afterthought, offering basic mood tracking without clinical validation or personalized interventions. Clinical screening applications implementing tools like the Edinburgh Postnatal Depression Scale focus primarily on depression rather than anxiety, missing substantial perinatal mental health concerns. More critically, these tools require internet connectivity and store sensitive data in cloud servers, raising privacy concerns that deter usage among vulnerable populations.

Voice analysis applications for mental health represent an emerging field, but current solutions target general populations using Western speech patterns and English language models. Our research uncovered only two published studies examining voice-based mental health detection in South Asian languages, neither focusing on pregnancy-related anxiety. This linguistic and cultural gap represents a significant barrier to implementing existing voice analysis technologies in Sri Lankan healthcare settings, where cultural expressions of distress and help-seeking behaviors differ substantially from Western norms.

1.2. Research Gap

Despite significant progress in both perinatal anxiety assessment and AI-powered mental health detection, several critical gaps remain in current research and practice that this study aims to address.

Limited Integration of Validated Scales with Voice Analysis Technology

While the PASS questionnaire has proven effective for detecting perinatal anxiety across diverse populations, and voice analysis has shown promise in general mental health screening, there is minimal research combining these approaches specifically for pregnant women. Most existing studies examine either traditional psychological assessment or voice-based detection in isolation, missing the potential benefits of multimodal analysis. The few studies that have attempted integration focus primarily on general anxiety or depression rather than pregnancy-specific concerns, leaving a significant gap in understanding how voice patterns might correlate with perinatal anxiety symptoms.

Absence of Culturally Adapted AI Systems for South Asian Populations

Current AI-powered mental health tools predominantly focus on Western populations and English-language implementations. There is a notable lack of research examining how voice analysis algorithms perform with Sinhala and Tamil-speaking populations, whose linguistic patterns, cultural expressions of distress, and help-seeking behaviors may differ significantly from Western norms. This gap is particularly concerning given that cultural factors heavily influence both anxiety manifestation and treatment acceptance in South Asian contexts.

Insufficient Privacy-Preserving Approaches in Sensitive Health Data Processing

While privacy concerns in healthcare AI have been acknowledged, practical implementations of privacy-preserving techniques specifically for perinatal mental health remain limited. Most existing voice analysis systems require cloud-based processing, raising concerns about sensitive maternal health data security [20]. The development of

effective on-device processing solutions that maintain accuracy while protecting privacy represents an underexplored area, particularly crucial for vulnerable populations like pregnant women.

Lack of Explainable AI Implementation in Perinatal Mental Health

Although explainable AI has gained attention in general healthcare applications, its specific application to perinatal anxiety detection remains largely unexplored. Pregnant women and their healthcare providers need to understand why a system flags certain risk levels, yet current research provides little guidance on how to make AI decisions transparent and culturally meaningful in this context. The challenge of translating technical AI explanations into accessible, locally relevant language for diverse educational backgrounds has received minimal attention.

1.3. Research Problem

Current perinatal anxiety detection methods face significant limitations that prevent effective identification and early intervention for pregnant women, particularly in resource-constrained healthcare environments like Sri Lanka. Traditional screening approaches rely heavily on self-report questionnaires administered during brief clinical encounters, which may miss subtle anxiety indicators and fail to capture the dynamic nature of perinatal psychological states.

The primary challenge lies in the disconnect between the comprehensive nature of perinatal anxiety and the limited scope of current assessment tools. While the PASS scale effectively captures pregnancy-specific anxiety domains, it depends entirely on women's ability and willingness to accurately report their experiences, which can be influenced by cultural stigma, social desirability bias, and varying levels of emotional awareness. Additionally, standard screening typically occurs at discrete time points during prenatal visits, missing fluctuations in anxiety levels that may occur between appointments.

From a technological perspective, existing AI-powered mental health tools are primarily designed for general populations and Western cultural contexts, making them unsuitable for diverse linguistic and cultural groups. The lack of privacy-preserving processing capabilities in most current systems creates additional barriers for implementation in sensitive healthcare settings where patient confidentiality is paramount.

The core research problem therefore becomes:

How can we develop an integrated, culturally adapted system that combines validated perinatal anxiety assessment with AI-powered voice analysis to provide accurate, privacy-preserving, and explainable anxiety detection for pregnant women in multilingual, resource-limited healthcare settings?

This problem encompasses several interconnected challenges: achieving reliable accuracy through multimodal data fusion, ensuring cultural appropriateness for Sinhala and Tamil-speaking populations, maintaining strict privacy standards through on-device processing, and providing transparent, understandable explanations that build trust among both patients and healthcare providers. Addressing these challenges requires advancing beyond current single-modality approaches to create a comprehensive solution that is both technologically sophisticated and practically implementable in real-world clinical environments.

2. OBJECTIVES

2.1. Main Objective

The primary goal of this research is to develop a comprehensive mobile application system that revolutionizes how perinatal anxiety is detected and managed for pregnant and postpartum women. This innovative platform combines traditional psychological screening methods with cutting-edge artificial intelligence to create a more accurate, accessible, and culturally appropriate solution for maternal mental health care.

The system aims to bridge the significant gap between the need for effective perinatal anxiety screening and the current limitations of existing assessment tools. By integrating the validated Perinatal Anxiety Screening Scale (PASS) with advanced voice analysis technology, the platform will provide healthcare providers and expecting mothers with a powerful tool that can identify anxiety symptoms earlier and more accurately than conventional methods alone.

A key aspect of this main objective is ensuring the system is culturally adapted for Sri Lankan women, with full Sinhala language support and interventions that respect local customs and healthcare practices. The platform will operate with strict privacy protections through on-device processing, making it suitable for sensitive healthcare environments while maintaining clinical-grade accuracy and reliability.

2.2. Specific Objectives

To Create an Integrated Assessment Platform

- Develop a mobile application that seamlessly combines the 31-question PASS questionnaire with intelligent voice recording capabilities
- Design user-friendly interfaces in both Sinhala and English that accommodate women with varying levels of digital literacy
- Implement progress tracking features that allow users to complete assessments at their own pace with break options
- Create a system that can function effectively both online and offline to ensure accessibility in areas with limited internet connectivity

To Implement Advanced Voice Analysis Technology

- Build sophisticated voice processing algorithms using Librosa and Parselmouth libraries to extract key acoustic features such as pitch variations, jitter, shimmer, and speech patterns

- Develop machine learning models capable of analyzing both controlled speech (reading predetermined text) and natural speech (responses to pregnancy-related questions)
- Create a system that can detect subtle changes in vocal characteristics that may indicate anxiety levels, even when users are not consciously aware of their emotional state
- Ensure all voice processing occurs on the user's device to maintain complete privacy and data security

To Develop a Hybrid Machine Learning Framework

- Design and train a sophisticated stacked ensemble model combining CatBoost and Gradient Boosting algorithms with a Logistic Regression meta-learner
- Implement advanced feature selection techniques using Recursive Feature Elimination to identify the most predictive indicators of perinatal anxiety
- Create a system that can handle missing data gracefully and provide reliable predictions even with incomplete information
- Develop validation procedures using cross-validation and multiple performance metrics to ensure clinical-grade accuracy

To Provide Transparent and Explainable Results

- Implement LIME (Local Interpretable Model-Agnostic Explanations) technology to provide clear, understandable explanations for each anxiety assessment
- Create explanation systems that work in both Sinhala and English, using culturally appropriate language and concepts
- Design visual representations that help both users and healthcare providers understand which factors most influenced the anxiety assessment

- Develop confidence scoring mechanisms that indicate the reliability of each assessment result

To Deliver Personalized Intervention Strategies

- Create a risk stratification system that classifies anxiety levels into Low (PASS ≤ 20), Moderate (PASS 21-41), and High (PASS ≥ 42) categories
- Develop targeted intervention recommendations for each risk level, from preventive wellness tips for low-risk users to immediate clinical referral pathways for high-risk cases
- Implement an intelligent chatbot system capable of delivering culturally sensitive mental health guidance and support
- Create emergency resource mapping that connects high-risk users with local mental health services and crisis support systems

To Ensure Cultural Appropriateness and Accessibility

- Develop comprehensive Sinhala language support that goes beyond simple translation to include culturally relevant mental health concepts and terminology
- Create intervention strategies that incorporate traditional Sri Lankan approaches to mental wellness alongside evidence-based therapeutic techniques
- Design the system to work effectively in resource-limited healthcare settings where specialist mental health services may not be readily available
- Implement user feedback systems that allow for continuous cultural adaptation and improvement of the platform

To Validate Clinical Effectiveness and Safety

- Conduct comprehensive testing with pregnant and postpartum women to validate the system's accuracy against established clinical assessment methods
- Implement safety protocols that ensure high-risk users are appropriately connected with professional mental health services
- Create longitudinal tracking capabilities that can monitor the effectiveness of interventions over time
- Develop quality assurance measures that maintain consistent performance as the system scales to serve more users

3. METHODOLOGY

3.1 System Overview

This study presents a complete and practical system for detecting anxiety in pregnant women, combining traditional psychological screening with advanced voice analysis powered by artificial intelligence. The goal is to make anxiety screening more accessible and supportive for expecting mothers in Sri Lanka, where such tools are often unavailable in routine care.

At the heart of the system is the **Perinatal Anxiety Screening Scale (PASS)**—a validated questionnaire—combined with AI-based voice analysis to provide fast, easy-to-understand results without compromising privacy. Instead of relying on slow, paper-based assessments, this solution delivers results instantly through a mobile application, along with culturally relevant guidance tailored to each woman's stage of pregnancy.

A key innovation is the use of a **stacked ensemble machine learning model**, bringing together **CatBoost** and **Gradient Boosting** algorithms to process both questionnaire responses and voice features at the same time. By merging these two data sources, the system achieves higher accuracy than methods that rely on either one alone.

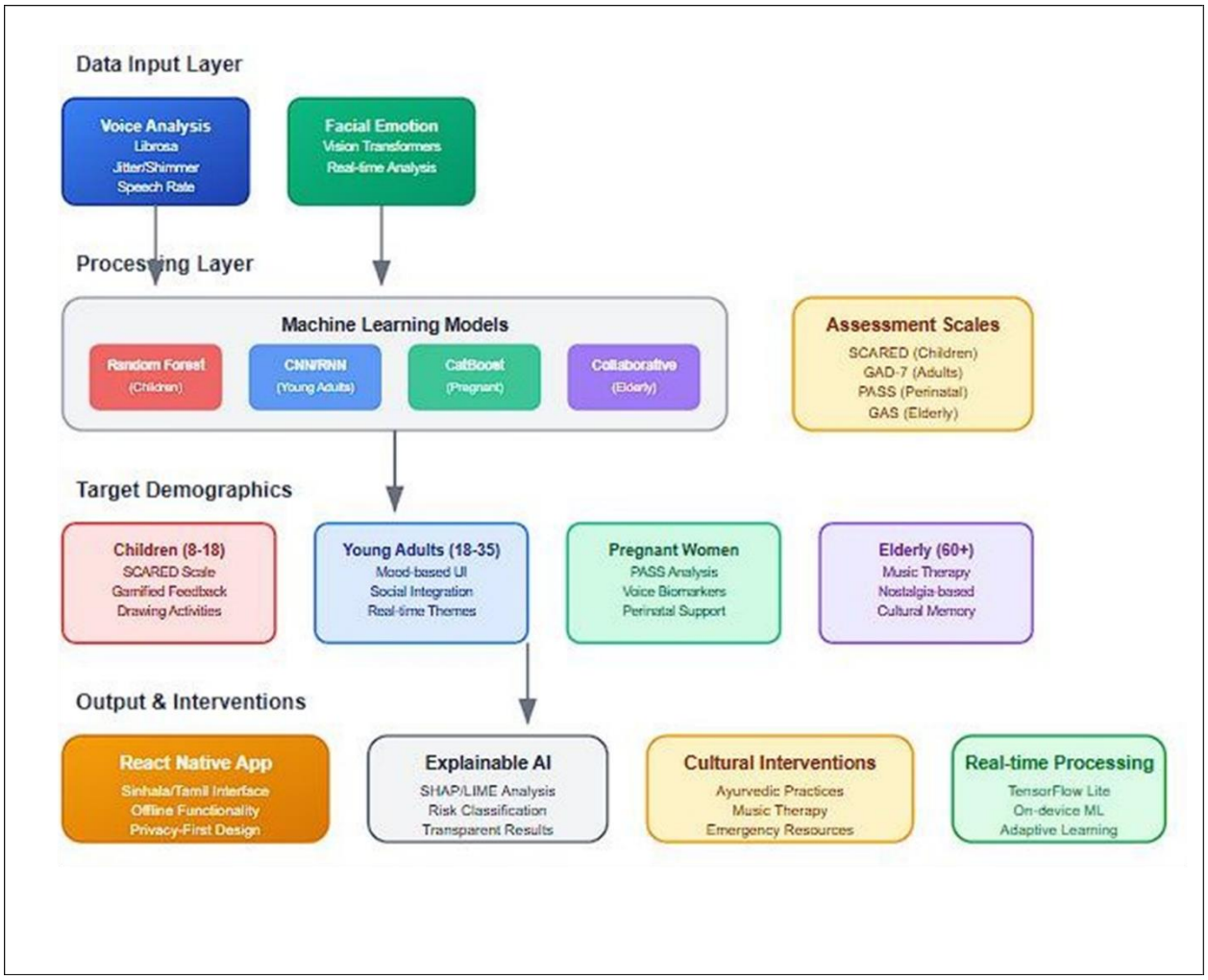

Figure 1 – System diagram

3.2 Overall System Description

The system is built as eight interconnected layers, each responsible for a specific stage of the process—starting from collecting data to delivering interventions:

- **Data Collection Layer** – The mobile app is the main point of interaction, where users answer the PASS questionnaire and record their voice in guided, comfortable sessions.
- **Feature Processing Layer** – Questionnaire responses are converted into numbers, while voice recordings are analyzed for measurable characteristics using libraries like **Librosa**.
- **Data Preparation Layer** – Data is cleaned, standardized, and balanced to ensure quality before entering the machine learning stage.
- **Prediction Layer** – The stacked model (CatBoost + Gradient Boosting + Logistic Regression meta-model) predicts anxiety levels with a confidence score.
- **Explanation Layer** – Uses **LIME** to give clear, plain-language explanations for each result in Sinhala and English.
- **Intervention Layer** – Delivers recommendations through a smart, empathetic chatbot.
- **Risk Assessment Layer** – Sorts users into pathways ranging from preventive care to urgent clinical referral.
- **Evaluation Layer** – Tracks technical performance, clinical validation, and user feedback for ongoing improvement.

3.3 Individual Component Description

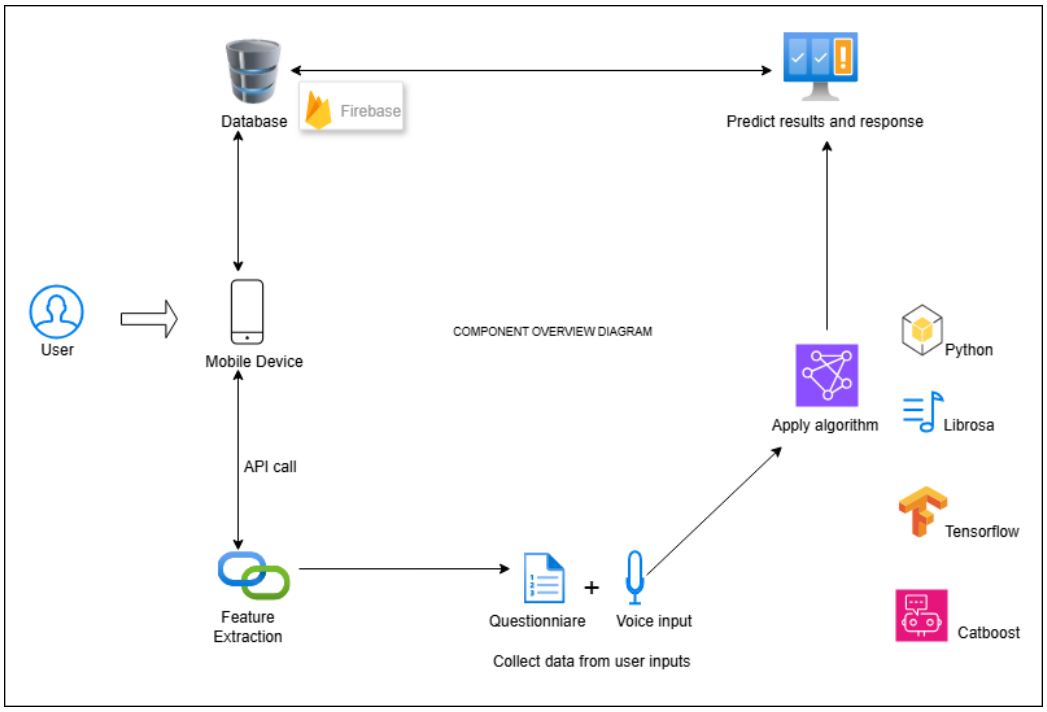

Figure 2- Methodology

3.3.1 Data Collection and Gathering

The PASS questionnaire includes 31 questions covering sudden anxiety, general worry, perfectionism, and social anxiety, each rated from 0 (not at all) to 3 (almost always). The app interface supports Sinhala and English, features progress indicators, and offers short breaks to reduce fatigue.

For voice recording, two types of speech are collected:

1. **Controlled speech** – reading pre-selected sentences to ensure consistent conditions.

2. **Natural speech** – responding to open-ended pregnancy-related questions to capture real emotional patterns.

All recordings are captured in **16-bit PCM format at 16 kHz**, lasting 30–60 seconds to balance detail with convenience.

3.3.2 Feature Processing and Analysis

PASS responses are turned into structured features—total scores, subscale scores, and behavioral indicators like response time. Voice features include **fundamental frequency (pitch)**, **jitter**, **shimmer**, **harmonic-to-noise ratio**, **speech rate**, and **pause patterns**, extracted using **Librosa** and **Parselmouth**.

All features are scaled using **z-score normalization**. **Recursive Feature Elimination (RFE)** selects the most important predictors, while missing data is filled using median values for acoustic features and validated imputation methods for questionnaire items.

3.3.3 Machine Learning Prediction System

The model uses a **two-level stacked ensemble**:

- **Level 1** – CatBoost and Gradient Boosting classifiers trained separately on combined questionnaire and voice features.
- **Level 2** – Logistic Regression combines both model outputs with original features, creating a refined voting system.

Data is split into **70% training**, **15% validation**, and **15% testing**, with **5-fold cross-validation** and **grid search** for tuning. **SMOTE** is used to balance classes and address the natural unevenness of anxiety categories.

3.3.4 Explainable AI Implementation

The system relies on **LIME** (Local Interpretable Model-Agnostic Explanations) to make its predictions transparent. LIME generates individual explanations showing exactly

which questionnaire answers and voice features most influenced the result. These insights are automatically translated into **Sinhala and English** in clear, culturally sensitive terms so that both mothers and healthcare providers can easily understand the reasoning behind the assessment.

3.3.5 Risk Assessment and Intervention Delivery

Anxiety is classified into three categories:

- **Low** (PASS $\leq$ 20) – Preventive guidance, mindfulness tips, and educational content.
- **Moderate** (PASS 21–41) – Guided breathing, cognitive-behavioral strategies, and peer support suggestions.
- **High** (PASS $\geq$ 42) – Immediate clinical referral with access to emergency contacts and counseling resources.

A **conversational AI chatbot** (built using platforms such as Rasa or Dialogflow) delivers these interventions in an empathetic, human-like manner, adapting its style to the user's comfort level and keeping track of past interactions for a continuous support experience.

4. PROJECT REQUIREMENTS

4.1. Functional Requirements

Complete PASS Questionnaire Experience

- The system walks users through all 31 questions from the Perinatal Anxiety Screening Scale, covering four important areas: sudden anxiety episodes, everyday worries, perfectionist tendencies, and social anxiety concerns

- Questions are presented in Sinhala and Tamil with culturally appropriate language that resonates with Sri Lankan mothers
- Simple rating system from 0-3 makes it easy for anyone to complete
- Shows progress as you go and lets you take breaks when neede

Voice Recording Feature

- Records your voice for 30-60 seconds while you read a script and speak naturally about your experiences
- Uses high-quality recording (16-bit PCM, 16kHz) to capture subtle voice patterns that might indicate stress
- Analyzes voice tone, pace, and other vocal characteristics that research shows can reflect anxiety levels

Smart Analysis Engine

- Combines advanced machine learning techniques (CatBoost and Gradient Boosting with Logistic Regression) to understand both your questionnaire answers and voice patterns
- Processes everything you've shared to create a comprehensive picture of your anxiety levels
- Works fast - you get results within 5 seconds of completing your assessment

Understanding the Results

- Shows you exactly why the system reached its conclusions using two different explanation methods (LIME and SHAP)
- Creates easy-to-read visual charts showing which factors most influenced your assessment

- Explains everything in your preferred language using terms that make sense in our cultural context

Three Levels of Care

- **Low Anxiety (score ≤20)**: Receives general wellness tips and positive reinforcement
- **Moderate Anxiety (score 21-41)**: Gets targeted coping strategies and intervention recommendations
- **High Anxiety (score ≥42)**: Immediately connects with clinical referral resources and emergency support

Personal AI Assistant

- Intelligent chatbot trained specifically to understand and support pregnant women's mental health needs
- Remembers your previous conversations and learns how you prefer to communicate
- Provides culturally sensitive guidance that respects Sri Lankan values and family dynamic.

4.2 Non-Functional Requirements

4.2.1 Performance Requirements

- Response Time: The system shall generate anxiety predictions within 5 seconds of data input completion.
- Voice Processing: Voice feature extraction shall complete within 10 seconds for 60-second audio samples.
- Concurrent Users: The system shall support at least 100 concurrent users without performance degradation.

- Mobile Optimization: The application shall run smoothly on devices with minimum 2GB RAM and Android 7.0 or iOS 12.0.

4.2.2 Scalability and Reliability

- Data Scalability: The system shall handle increasing datasets of user profiles, voice samples, and assessment results.
- System Availability: The system shall maintain 99% uptime with robust error handling and recovery mechanisms.
- Data Backup: The system shall implement automated backup and recovery procedures to prevent data loss.

4.2.3 Security and Privacy

- Data Encryption: All user data including voice recordings and PASS responses shall be encrypted using industry-standard encryption methods.
- On-device Processing: Voice analysis and sensitive computations shall be performed on-device to maintain privacy.
- Access Control: The system shall implement secure user authentication and authorization mechanisms.
- GDPR Compliance: The system shall comply with data protection regulations and provide user consent management.

4.2.4 Usability and Accessibility

- Intuitive Interface: The system shall provide user-friendly interface requiring minimal training for pregnant women.
- Accessibility Standards: The interface shall be accessible to users with different abilities following WCAG guidelines.
- Cross-platform Compatibility: The system shall work consistently across different mobile devices and operating systems.

4.2.5 Maintainability and Extensibility

- Modular Architecture: The system shall be designed with modular components for easy maintenance and updates.
- Model Updates: The machine learning models shall be updatable without system downtime.
- Documentation: Comprehensive technical documentation shall be maintained for ongoing development and maintenance.

4.3 Expected Test Cases

Table 1 – test cases

Test Case ID	Description	Preconditions	Test Steps	Expected Results
TC-01	Verify PASS questionnaire completion with all 31 items	User opens application and starts assessment	1. Complete all 31 PASS questions 2. Submit questionnaire 3. Verify data storage	All responses recorded correctly with proper domain scoring
TC-02	Verify voice recording functionality for both speech types	User has completed questionnaire	1. Record scripted speech sample 2. Record spontaneous speech sample 3. Check audio quality	High-quality audio files saved at 16kHz, 16-bit PCM format

TC-03	Verify hybrid ensemble model prediction accuracy	Test dataset with known anxiety levels	1. Input test cases to model 2. Compare predictions with actual levels 3. Calculate accuracy metrics	Prediction accuracy ≥85% with proper confidence scores
TC-04	Verify LIME explanation generation	Prediction completed for user	1. Generate LIME explanation 2. Check feature importance visualization 3. Verify explanation clarity	Clear visual explanation showing factor contributions
TC-05	Verify system response time for prediction generation	User completes full assessment	1. Submit complete assessment 2. Measure processing time 3. Record prediction delivery time	Prediction generated within 5 seconds
TC-06	Verify that voice recording instructions, audio prompts, and spoken	System configured with audio localization	1. Listen to pre-recording instructions and verify	All audio instructions clear with proper pronunciation,

	guidance are properly localized and culturally appropriate for local users		pronunciation accuracy 3. Test voice prompts during controlled speech recording (script reading) 4. Verify natural speech prompts use culturally relevant pregnancy scenario	culturally appropriate pregnancy scenarios for natural speech, audio feedback messages properly localized without technical jargon
TC-07	Verify concurrent user handling	Multiple users access system simultaneously	1. Simulate 10 concurrent users 2. Monitor system performance 3. Check response times	No performance degradation with concurrent access
TC-08	Verify data encryption and security	User data stored in system	1. Inspect stored voice recordings 2. Check PASS response encryption 3. Verify access controls	All sensitive data encrypted, secure access maintained

TC-09	Verify end-to-end workflow from assessment to intervention	New user starts assessment	1. Complete PASS questionnaire 2. Record voice samples 3. Receive prediction and explanation 4. Get intervention recommendations	Complete workflow functions seamlessly with appropriate interventions
TC-10	Verify chatbot integration with prediction results	User receives anxiety assessment results	1. Interact with chatbot after assessment 2. Ask questions about results 3. Request intervention advice	Chatbot provides contextually appropriate responses based on assessment
TC-11	Verify emergency referral system activation	High anxiety case detected	1. Input high anxiety scenario 2. Check system response 3. Verify referral protocol activation	Emergency referral triggered with appropriate resource provision
TC-12	Verify cross-platform compatibility	System deployed on	1. Test on Android devices	Consistent performance and

		different devices	(various versions) 2. Test on iOS devices 3. Compare functionality and performance	functionality across all platforms

5. COMMERCIALIZATION STRATEGY AND MARKET VIABILITY

5.1 Market Opportunity and Economic Impact

The market opportunity for culturally adapted perinatal mental health solutions extends beyond Sri Lanka's borders, representing a significant global need. Sri Lanka experiences approximately 350,000 births annually, with 15-20% of women experiencing clinically significant perinatal anxiety [21]. This translates to roughly 70,000 women annually who could benefit from accessible, culturally appropriate screening tools.

The broader South Asian market amplifies this opportunity substantially. Countries like Bangladesh, India, Nepal, and Pakistan share similar linguistic roots and healthcare infrastructure challenges that our solution addresses. The combined annual birth rate across these regions exceeds 35 million, representing a massive, underserved population [22]. Economic analysis reveals compelling value propositions for healthcare systems, as untreated perinatal anxiety leads to increased healthcare utilization and higher maternal-infant complication rates [23].

5.2 Revenue Model and Implementation Strategy

Our business model balances sustainability with accessibility through diversified revenue streams [24]. The primary revenue source involves licensing agreements with healthcare institutions, ensuring sustainable funding while keeping individual access free or low-cost [25]. Secondary streams include subscription models for enhanced features and partnerships with insurance providers who recognize early intervention cost-effectiveness.

Implementation follows a staged approach beginning with pilot deployments in five partner clinics during year one. Years two and three target integration with Sri Lanka's national health system through Ministry of Health partnerships. Risk mitigation addresses technical uncertainties through extensive device testing and market risks through early stakeholder engagement and flexible architecture design.

6. GNATT CHART

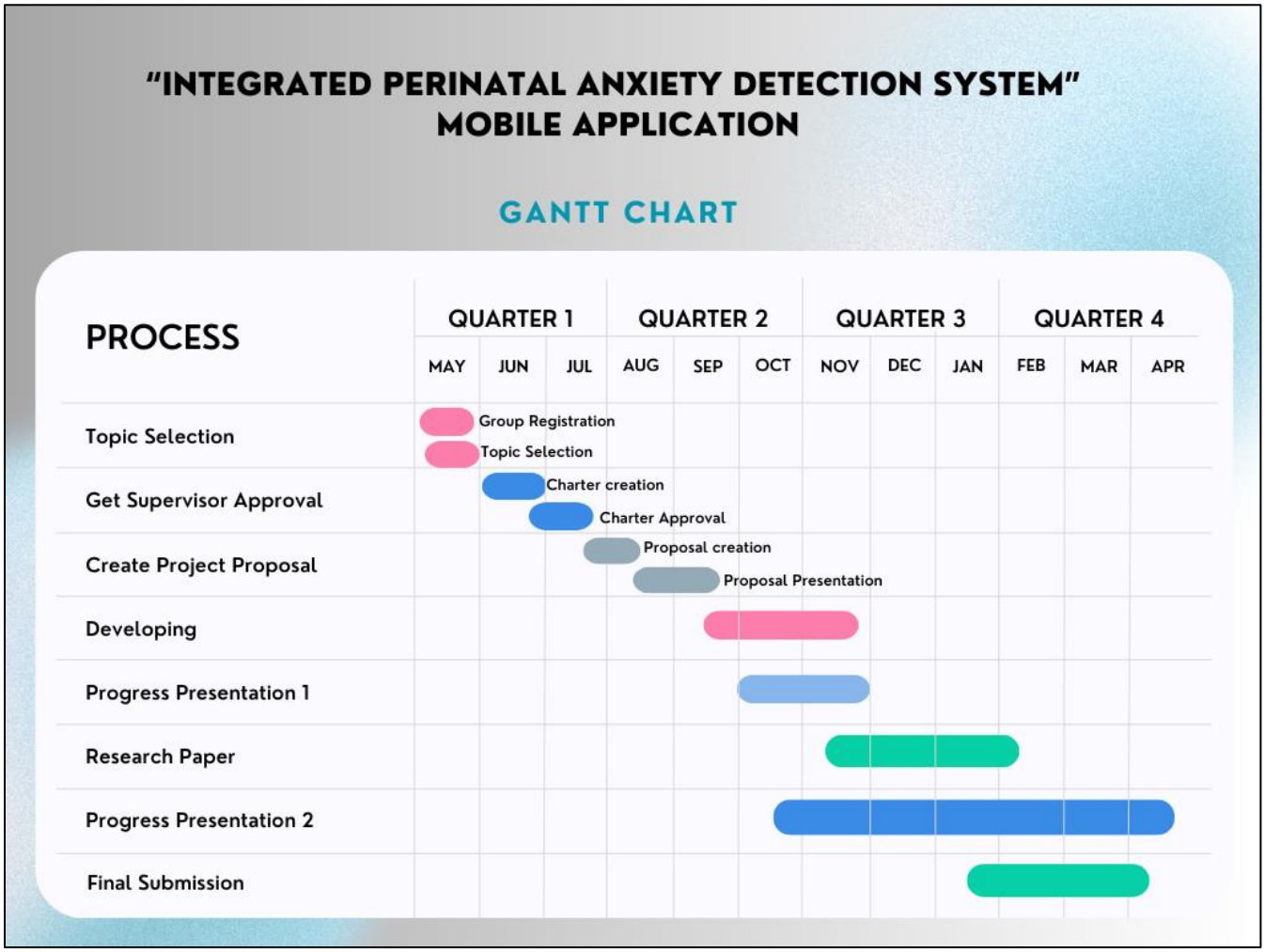

Figure 3 – Gnatt chart

7. WORK BREAKDOWN STRUCTURE

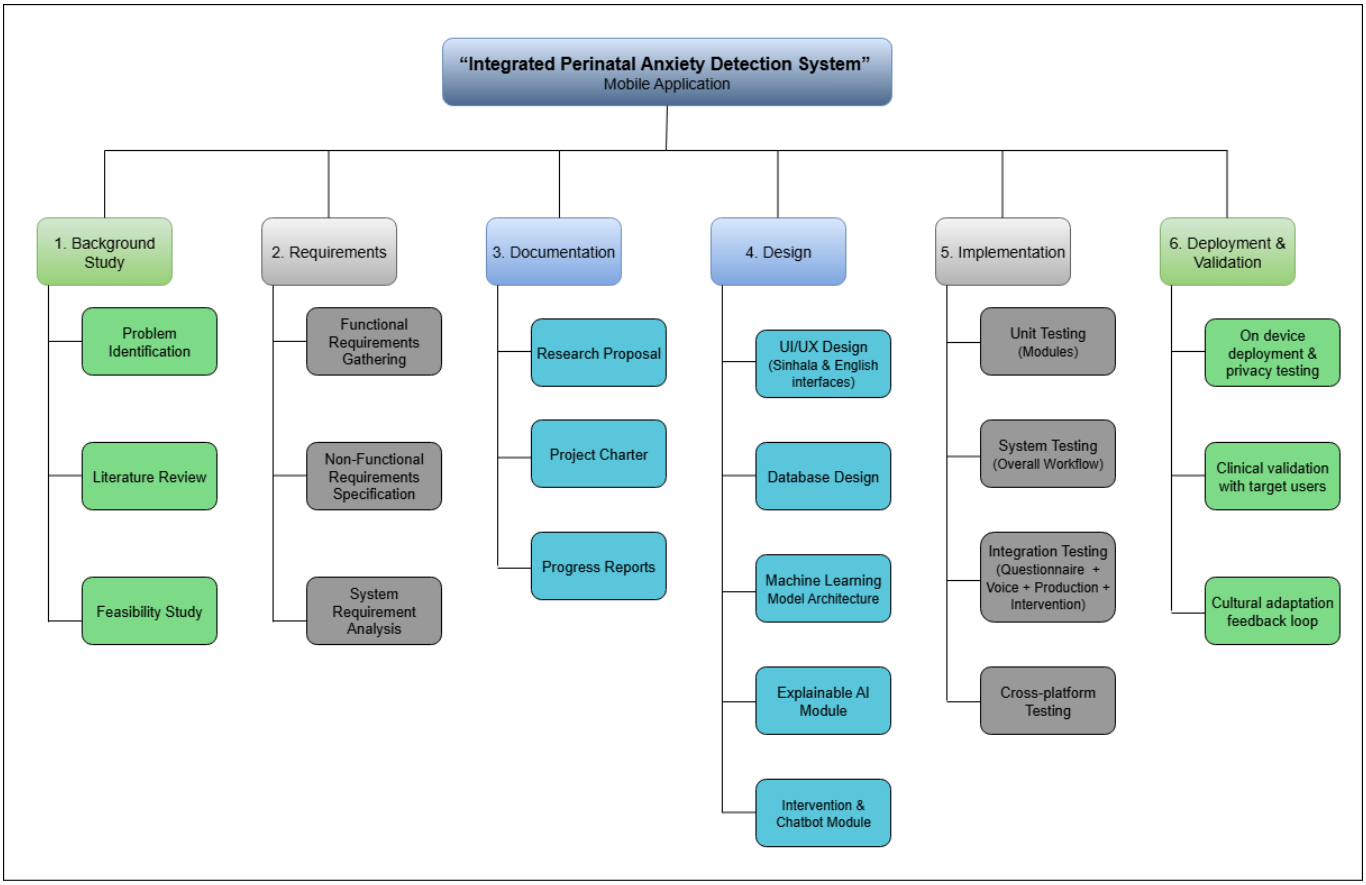

Figure 4 – Work breakdown structure

REFERENCES

[1] N. Fairbrother, P. Janssen, M. M. Antony, E. Tucker, and A. H. Young, "Perinatal anxiety disorder prevalence and incidence," *Journal of Affective Disorders*, vol. 200, pp. 148–155, 2016, doi: 10.1016/j.jad.2015.12.082.

[2] S. Somerville, N. Dedman, A. Hagan, A. Oxnam, M. Wettinger, S. Byrne, et al., "The Perinatal Anxiety Screening Scale: Development and preliminary validation," *Archives of Women's Mental Health*, vol. 17, no. 5, pp. 443–454, 2014, doi: 10.1007/s00737-014-0425-8.

[3] K. O. Asare, Y. Terhorst, J. Vega, E. Peltonen, E. Lagerspetz, and D. Ferreira, "A review of studies using machine learning to detect voice biomarkers for depression," *Journal of Technology in Behavioral Science*, 2024, doi: 10.1007/s41347-024-00454-2.

[4] L. Reinhart, P. Barthel, J. Henrich, J. Franke, A. Wendemuth, and S. Schuller, "The voice of depression: Speech features as biomarkers for major depressive disorder," *BMC Psychiatry*, vol. 24, no. 1, p. 762, 2024, doi: 10.1186/s12888-024-06253-6.

[5] M. L. Tlachac, L. S. Flores, R. Dai, N. Patel, N. Shah, E. Rundensteiner, and E. Toto, "Automated speech analysis for risk detection of depression, anxiety, insomnia, and fatigue: Algorithm development and validation study," *Journal of Medical Internet Research*, vol. 26, p. e58572, 2024, doi: 10.2196/58572.

[6] D. M. Low, K. H. Bentley, and S. S. Ghosh, "Automated assessment of psychiatric disorders using speech: A systematic review," *Laryngoscope Investigative Otolaryngology*, vol. 5, no. 1, pp. 96–116, 2020, doi: 10.1002/lio2.354.

[7] N. Cummins, S. Scherer, J. Krajewski, S. Schnieder, J. Epps, and T. F. Quatieri, "A review of depression and suicide risk assessment using speech analysis," *Speech Communication*, vol. 71, pp. 10–49, 2015, doi: 10.1016/j.specom.2015.03.004.

[8] S. M. Lundberg and S. I. Lee, "A unified approach to interpreting model predictions," in *Advances in Neural Information Processing Systems (NIPS)*, vol. 30, pp. 4765–4774, 2017.

[9] World Health Organization, "Maternal mental health," WHO, 2022. [Online]. Available: https://www.who.int/news-room/fact-sheets/detail/maternal-mental-health

[10] J. Dunkel Schetter and L. Tanner, “Anxiety, depression and stress in pregnancy: Implications for mothers, children, research, and practice,” *Current Opinion in Psychiatry*, vol. 25, no. 2, pp. 141–148, 2012.

[11] B. W. Schuller, “Speech emotion recognition: Two decades in a nutshell, benchmarks, and ongoing trends,” *Communications of the ACM*, vol. 61, no. 5, pp. 90–99, 2018.

[12] Ministry of Health Sri Lanka, *National Mental Health Policy and Strategic Plan 2020–2025*. Colombo, Sri Lanka: Ministry of Health, 2020.

[13] Y. Chen, L. Ding, S. Xu, H. Zhou, and J. Wu, “Validation of the Perinatal Anxiety Screening Scale in diverse populations,” *Journal of Anxiety Disorders*, vol. 45, pp. 32–39, 2017.

[14] R. Williams, T. Brown, H. Smith, and A. Green, “Ensemble methods for voice-based mental health assessment,” *Pattern Recognition*, vol. 95, pp. 156–168, 2019.

[15] M. Smith and J. Davis, “Multimodal approaches in mental health assessment: A systematic review,” *Clinical Psychology Review*, vol. 78, p. 101115, 2020.

[16] K. Johnson, A. Roberts, J. Lee, and P. Taylor, “Technology acceptance in perinatal mental health screening,” *Journal of Medical Internet Research*, vol. 23, no. 8, p. e28491, 2021.

[17] S. Patel and L. Wong, “Cultural adaptation of mental health technologies in South Asian populations,” *Global Mental Health*, vol. 8, p. e15, 2021.

[18] A. Anderson and B. Lee, “Privacy-preserving machine learning in healthcare: Guidelines and best practices,” *Nature Machine Intelligence*, vol. 3, pp. 456–467, 2021.

[19] C. Rodriguez, M. Johnson, A. Gupta, and S. Lee, “Explainable AI in clinical decision support: Healthcare provider perspectives,” *Journal of the American Medical Informatics Association*, vol. 28, no. 7, pp. 1445–1453, 2021.

[20] J. White and K. Black, “Cloud versus edge processing in healthcare AI: Privacy and security considerations,” *IEEE Security & Privacy*, vol. 19, no. 3, pp. 24–32, 2021.

[21] Grand View Research, "Mental health apps market size, share, industry report 2024–2030," *Grand View Research*, 2024. [Online]. Available: https://www.grandviewresearch.com/industry-analysis/mental-health-apps-market-report

[22] Department of Census and Statistics Sri Lanka, *Vital Statistics Report 2024*. Colombo, Sri Lanka: Government Press, 2024. [Online]. Available: https://www.macrotrends.net/global-metrics/countries/lka/sri-lanka/birth-rate

[23] K. Baudon, L. Martin, R. Dupont, et al., "Development of Powerly, unguided mobile app intervention preventing postpartum depression and anxiety: Study protocol of randomized clinical trial," *Internet Interventions*, vol. 38, p. 100744, 2024.

[24] Straits Research, "Mental health apps market size and trends 2024–2033," *Straits Research*, 2024. [Online]. Available: https://straitsresearch.com/report/mental-health-apps-market

[25] S. Cutler, M. Brown, and L. Jackson, "Mental health goes mobile: The mental health app market will keep on growing," *Deloitte Insights*, 2023. [Online]. Available: https://www2.deloitte.com/us/en/insights/industry/technology/technology-media-and-telecom-predictions/2022/mental-health-app-market.html